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Investigating Locomotion Techniques for Blind People in Virtual Reality

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[1] Tech & People Team - `https://techandpeople.github.io/team`
[2] AccessVR Project - `https://doi.org/10.54499/2022.08286.PTDC`

Resumo

A evolução das interações tradicionais para abordagens mais naturais e inclusivas em ambientes virtuais, tem sido marcada por avanços significativos em termos de tecnologia, design e acessibilidade. Esta evolução caracteriza-se por uma transição de interfaces estáticas (e.g. teclado, rato) e frequentemente exclusivas, para métodos dinâmicos e intuitivos (e.g. gestos, comandos de voz) que respondem a uma vasta gama de necessidades dos utilizadores. A Realidade Virtual (RV) tira partido destas modalidades naturais, transformando a forma como interagimos com os ambientes virtuais [1]. A natureza imersiva da RV permite aos utilizadores explorar e navegar em mundos virtuais, tornando as técnicas de locomoção um aspeto crucial da experiência de RV. Estas técnicas permitem aos utilizadores interagir com os ambientes virtuais de formas anteriormente impossíveis, como andar, correr ou saltar fisicamente.

Investigadores estudam extensivamente as numerosas técnicas de locomoção propostas e implementadas em contextos comerciais e académicos. Muitas revisões da literatura, taxonomias e bases de dados (e.g., [1, 2, 3]) tentaram definir as técnicas de locomoção existentes e estudar elementos essenciais como a imersão, o cansaço, o enjoo, entre outros. As revisões da literatura atribuem frequentemente às técnicas de locomoção categorias específicas, criando taxonomias para agrupar técnicas de locomoção semelhantes. Ainda assim, as revisões e as comparações existentes entre um subconjunto de técnicas mostram os prós e os contras de cada técnica, por vezes com conclusões contraditórias. Estas diferenças sustentam a necessidade de investigar técnicas proeminentes em diferentes cenários. No entanto, a acessibilidade das técnicas de locomoção não é abordada nas comparações ou revisões da literatura, apesar dos desafios conhecidos que as pessoas com deficiência enfrentam com as tecnologias de RV [4]. Contudo, na literatura existe uma exceção [1], uma base de dados com visualização (imagens e vídeos) que consolida informações sobre 109 técnicas de locomoção. Apesar do passo em frente, a acessibilidade é definida em termos da capacidade motora necessária para utilizar as técnicas, o que se deve provavelmente à investigação anterior sobre RV para pessoas com deficiências motoras (por exemplo, [5]). Mesmo assim, o conhecimento e abordagens sobre a acessibilidade das técnicas de locomoção para pessoas cegas é muito limitado.

Uma abordagem existente consiste em andar no mundo real, produzindo o mesmo efeito no mundo virtual. Esta abordagem foi utilizada em trabalhos anteriores no treino de orientação e mobilidade [6] e sobre a promoção da atividade física [7]. No entanto, a equivalência ao andar no mundo real impõe restrições, uma vez que a experiência tem de ser concebida para espaços

pequenos ou é necessário um grande espaço para utilizar o dispositivo de RV. Uma forma de lidar com as restrições de espaço físico é utilizar técnicas que permitem o utilizador andar no mesmo lugar. Outras técnicas, concebidas exclusivamente para pessoas cegas, utilizam as suas capacidades de mobilidade e ajudas à navegação para apoiar a mobilidade virtual (e.g. bengala aumentada com um sistema háptico [8]). No entanto, isto representa um pequeno subconjunto de técnicas de locomoção disponíveis, particularmente as disponíveis (e mais prevalentes) em experiências de RV comuns, que podem fornecer recursos para além da locomoção no mundo real. A este respeito, existe pouco ou nenhum conhecimento sobre a forma como diferentes técnicas de locomoção podem apoiar experiências de RV acessíveis para pessoas cegas e os seus níveis de conforto, facilidade de utilização e envolvimento.

Para obter mais conhecimento sobre a acessibilidade das técnicas de locomoção em RV, exploramos a base de dados previamente mencionada [1]. Desta forma, fomos observar a base de dados mais a fundo com base em diretrizes orientadas na acessibilidade (e.g. videojogos), em que no final propomos novas categorias, mas orientadas à acessibilidade. Dois investigadores adicionais foram introduzidos no estudo, em que no final catalogamos 107 técnicas de locomoção na tentativa de atingir um conjunto de deficiências. A catalogação baseou-se nos conteúdos registados na base de dados, se disponíveis (e.g., descrição das técnicas, e vídeos) e, caso não disponíveis, os vídeos foram pesquisados manualmente (e.g. nome do criador, registado na base de dados). Mostrando no final os nossos resultados.

De seguida, para ajudar a selecionar técnicas de locomoção em RV com o objetivo de investigar com mais detalhe a acessibilidade para pessoas cegas, analisamos as 107 técnicas de locomoção com as novas categorias propostas no primeiro estudo e com os atributos de experiência de utilização propostos na base de dados. Dito isto, procedemos a um processo de filtragem com foco na propriocepção para ver quais as que têm mais potencial para serem acessíveis. Contudo, também observamos quais as técnicas mais comuns no mercado, e prevalentes na literatura. No final, selecionamos três técnicas comuns, mas diferentes entre elas - "*Movimento de Braços*", "*Joystick*" e "*Teletransporte*". Nós adaptamos cada técnica, quando necessário, com pistas hápticas e auditivas para garantir a acessibilidade (por exemplo, o som de passos e o som háptico/som de colisões), tal como sugerido por trabalhos anteriores (e.g. [9]). Adicionalmente implementamos dois ambientes virtuais: um de aprendizagem para introduzir os participantes às técnicas e à aplicação de RV, e um ambiente de estudo para realizar as tarefas de estudo. As técnicas e ambientes, foram ambos implementados no Unity3D com os Meta Quest 2 e utilizamos sempre que pudéssemos os recursos do Unity3D. Realizámos um estudo com 14 participantes cegos que consistiu em tarefas de navegação com estas técnicas de locomoção e uma entrevista semi-estruturada, para responder a duas perguntas de investigação:

1. Quais são as diferenças entre "*Movimento de Braços*", "*Joystick*" e "*Teletransporte*" em termos de desempenho e preferências do utilizador?

2. Quais são as vantagens relativas de cada técnica e qual o impacto das interações e estratégias dos utilizadores no desempenho?

O estudo teve um design dentro-do-sujeito, em que as sessões duraram em média 84 minutos (mínimo 69 minutos, máximo 112 minutos) e todos os participantes executaram uma tarefa de navegação com três condições diferentes (as três técnicas de locomoção), em que a ordem de cada condição foi contrabalançada. A nossa metodologia gira em torno da análise quantitativa, envolvendo o desempenho dos participantes na tarefa e classificações auto-relatadas (por exemplo, conforto, diversão, eficiência), e da análise qualitativa baseada em entrevistas semi-estruturadas e observações.

Não encontrámos diferenças no desempenho geral, mas sim preferências contrastantes. Os resultados destacam os desafios e as vantagens de cada técnica e as estratégias utilizadas pelos participantes. Discutimos, entre outros aspetos, a forma como a adaptação das técnicas permitiu às pessoas cegas navegar em RV, o maior controlo do movimento do "*Movimento de Braços*", a simplicidade e familiaridade do "*Joystick*" e o potencial de eficácia do "*Teletransporte*".

Em suma, esta dissertação tem como principais contribuições:

- Expandir o conhecimento sobre a acessibilidade das técnicas de locomoção em RV, analisando e alargando o catálogo do *Locomotion Vault* [1]. Em particular, adicionando categorias relacionadas com a acessibilidade, como por exemplo, deficiências (por exemplo, Motor, Visão) e outras categorias orientadas para a acessibilidade (por exemplo, Modalidades, Inputs Alternativos).

- Uma análise alargada do catálogo para avaliar o potencial das técnicas de locomoção para apoiar a navegação de pessoas cegas em RV.

- Implementação de um ambiente virtual e de três técnicas de locomoção populares – "*Movimento de Braços*", "*Joystick*" e "*Teletransporte*" - aumentadas com feedback auditivo e outros feedbacks para as tornar acessíveis a pessoas cegas.

- A análise de um estudo de utilizador com 14 participantes, onde as três condições foram comparadas em termos de desempenho e preferência do utilizador. E, uma análise mais profunda para compreender a lógica subjacente às preferências e estratégias que afetam o desempenho.

Palavras-chave: Técnicas de Locomoção; Ambientes Virtuais; Realidade Virtual; Pessoas Cegas; Navegação.

Abstract

Locomotion plays a key role in Virtual Reality (VR), allowing users to navigate and explore the virtual world. Many VR locomotion techniques have been proposed, but those explored for and with blind people are often custom-made or require specialized equipment. Consequently, it is unclear how popular techniques can support blind people's VR locomotion, blocking access to most VR experiences. To gain more insight into the accessibility of VR locomotion techniques, we explored a database of 109 techniques. However, the focus is on the motor skills needed to use the techniques. Thus, we took a closer look at accessibility-oriented guidelines, proposing new categories to the database regarding accessibility. Then, to help select VR locomotion techniques with the aim of investigating accessibility for blind people in more detail, we analyzed the techniques with the new categories proposed in the first study, user experience attributes exposed in the database, literature and resources available on the market. We selected three common techniques - Arm Swinging, Linear Movement (joystick-based steering), and Point & Teleport - which are also very different among themselves. Then, we implemented and augmented them with audio and haptic cues for accessibility. We conducted a study with 14 blind participants consisting of navigation tasks with these techniques and a semi-structured interview. We found no differences in overall performance (e.g., completion time), but contrasting preferences. Findings highlight the challenges and advantages of each technique and strategies employed by participants. We discuss, among others, how augmenting the techniques enabled blind people to navigate in VR, the greater control of the movement of Arm Swinging, the simplicity and familiarity of Linear Movement, and the potential for efficiency of Point & Teleport.

Keywords: Locomotion Techniques; Virtual Environments; Virtual Reality; Blind People; Navigation.

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Acronyms

CAOTS Controller Assisted On The Spot.

HCI Human-Computer Interaction.

HMD Head-Mounted Display.

LTs Locomotion Techniques.

O&M Orientation and Mobility.

SDK Software Development Kit.

TTS Text-To-Speech.

UX User eXperience.

VEs Virtual Environments.

VR Virtual Reality.

W3C World Wide Web Consortium.

WHO World Health Organization.

XR Mixed Reality.

Chapter 1

Introduction

Visual impairments are surprisingly common, the World Health Organization (WHO) estimates that 2.2 billion people globally have a vision impairment [10]. While the exact number of blind people is unknown, a significant percentage of blind people in the world face major barriers navigating in the virtual world [9, 11].

1.1 Motivation

The interest in virtual environments using Virtual Reality (VR) is rapidly expanding due to recent technological advances. Locomotion plays a key role in VR, allowing users to navigate and explore the virtual world. Multiple techniques have been proposed and implemented [1, 2, 12], ranging from familiar digital interactions (e.g., gamepad thumbstick) to magical (e.g., teleporting) options or mimicking real-world movement (e.g, a treadmill or arm swing). Such a variety of locomotion techniques provides a range of experiences with different levels of immersion, comfort, and freedom of movement [1, 3]. In addition, the different interaction mechanisms satisfy different abilities (e.g., using the controllers, legs, or even posture to walk), potentially increasing the accessibility of VR experiences. Yet, the current focus is on sighted people's perspectives, since VR applications rely heavily on visual feedback, leaving little to no knowledge about the accessibility of locomotion techniques, particularly for blind and visually impaired people. Therefore, blind and visually impaired people face major difficulties interacting with current VR experiences. In particular, becoming aware of their surroundings and navigating virtual environments.

Prior research in accessible VR for blind people has focused on developing and assessing custom-made applications, for instance, to support Orientation and Mobility (O&M) training [13, 14, 15, 16]. However, this represents a small subset of available locomotion techniques, particularly those available (and more prevalent) in common VR experiences, which may provide affordances beyond real-world locomotion. In that regard, there is little to no knowledge of how different locomotion techniques may support accessible VR experiences for blind people and their levels of fun, comfort, and ease of use. Knowing the challenges, we want to understand if existing techniques are (or could be) accessible to blind people and give them access to a wider range of VR experiences.

1.2 Goals

Our focus was on supporting navigation in Virtual Environments (VEs), and for this, we investigated Locomotion Techniques (LTs) in VR to understand the abilities required to interact with each technique and see how to make them accessible for blind people. Our research approach started with an analysis of an existent database of 109 VR locomotion techniques [1]. We included categories related to accessibility, as a way to understand the abilities required to interact with each locomotion technique and, consequently, its potential to support accessible VR experiences. After that, to help select VR locomotion techniques with the aim of investigating accessibility for blind people in more detail, we analyzed the techniques with the new categories proposed in the first study, user experience attributes exposed in the database, literature, and resources available on the market. We have selected three common techniques but different among themselves, while also being popular in the literature and commercial VR applications. Then, we conducted a user study with 14 legally blind participants, in which we implemented the three techniques - Arm Swinging, Linear Movement (joystick-based steering), and Point & Teleport - with accessibility adaptations, to understand how these techniques support, or not, blind people navigating VR environments.

1.3 Contributions

Our main contributions to this thesis are:

- Expand the knowledge of the accessibility of VR locomotion techniques, by analysing and augmenting the *Locomotion Vault* [1]. In particular, adding the following accessibility-related categories: user requirements (e.g. Motor, Vision) and other accessible-driven categories (e.g. Input Modalities, Alternative Inputs).

- An extended analysis of the database to assess the potential of the locomotion techniques to support the navigation of blind people in VR.

- Implementation of a virtual environment and the three popular locomotion techniques - Arm Swinging, Linear Movement, and Point & Teleport - augmented with auditory and other feedback cues to make them accessible to blind people.

- The analysis of a user study with 14 participants, where the three conditions were compared in terms of performance and user preference. And, a deeper analysis to understand the rationale behind the preferences and strategies impact performance.

1.4 Publications

The results of the second study were accepted as a full paper to the *CHI'2024* conference:

[1] The Locomotion Vault - `https://locomotionvault.github.io/` (Last visited on April 12th, 2023).

Renato Ribeiro, Inês Gonçalves, Manuel Piçarra, Letícia Seixas Pereira, Carlos Duarte, André Rodrigues, João Guerreiro. 2024. *Investigating Locomotion in Virtual Reality with Blind People*. In Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (Hawaii, USA) (CHI '24). Association for Computing Machinery.

1.5 Statement of Collaboration

While the abovementioned section highlights my work, it is essential to acknowledge the collaboration of these endeavours. I played a central role in formulating concepts, implementing the prototype, conducting research, and manuscript development. My collaborators provided valuable support, enhancing the quality of the work. In particular, collaboration was vital when augmenting the *Locomotion Vault* [1] (bringing in varied perspectives); executing the user studies and discussing the different steps of this research.

1.6 Structure of the document

This document is organized as follows:

- **Chapter 2 - Related Work**. Overview of the state-of-the-art using virtual environments, the benefits and differences between types of interaction with virtual environments, and locomotion in virtual reality, specifically to blind people.
- **Chapter 3 - Cataloging the Accessibility of VR Locomotion Techniques**. Includes the rationale behind the proposed categories, our findings, and extending our analysis further to understand potential techniques for blind people.
- **Chapter 4 - Implementation of VR Locomotion Techniques**. The details regarding the selected locomotion techniques, environments, tools, and assets used to implement the VR prototype.
- **Chapter 5 - User Study**. Addresses the user study with 14 blind participants testing our environment and selected techniques to understand how the techniques could support them in navigating VR environments.
- **Chapter 6 - Conclusion**. Final thoughts on the current work and prospects of future work.

Chapter 2

Related Work

In this section, we discuss 1) the benefits, different interactions, and research efforts for accessible virtual environments for blind people, 2) the numerous locomotion techniques implemented and studied with sighted people, and 3) locomotion for blind people in VR.

2.1 Virtual Environments as Solutions

Virtual Environments (VEs) are digital spaces that enhance the interaction between humans and devices. The main advantage of using VEs is not being bound to the physical laws of the real world – it allows the blind person to train without fear, with no time restrictions - can pause, rewind, save progress, start in any location within the digital world, and use it independently – without a specialist. The information is received in feedback from the devices or peripherals that the blind person uses to navigate in the virtual environment. This feedback can be controlled, in it (VEs), is possible to design environments to communicate with a single sense (audio or haptic) or multiple senses (two or more). In addition, it allows data retrieval - to analyze the patterns, behaviours, strategies, and sensory stimulation perceived by blind people [17].

However, there are also disadvantages to the use of virtual environments. For instance, in mobility and orientation training the pre-planning aid's main purpose is to transfer the skills and strategies acquired in the virtual to the real world, but the VEs are not the exact model or substitute for the actual environment - this problem can be argued that comes mainly from the poor resolution or scale of the provided digital space [18]. Typically, virtual environments use a microscale framework – a space that does not require full-body movement [19], the lack of visual content of the space, and little body locomotion – cuts part of the sensorial information and can bring misunderstandings about the spatial information on the environment. Nonetheless, there are benefits to using the digital world that we can retain, like – the integration of the blind person; the engagement the blind person feels while interacting with the devices; the feedback received (getting an immediate response); and more importantly a space to try and develop tools in a trial-and-error manner without the dangerous settings in the real world.

2.2 Virtual Environment Interactions

The accessibility of virtual environments is frequently ignored, which is especially relevant for blind people, as interaction usually relies on visual stimuli. As such, *Giudice* quotes as follows, "*we should not approach it as loss of vision, we need to think about vision loss as a being use of a different state space of nonvisual information*" [20]. In context, the available cases in the nonvisual state are the remaining senses (e.g. hearing, touch, smell, and taste), but typically, in most VEs, only two senses are used to get information - **hearing (audio)** and **touch (haptic)**.

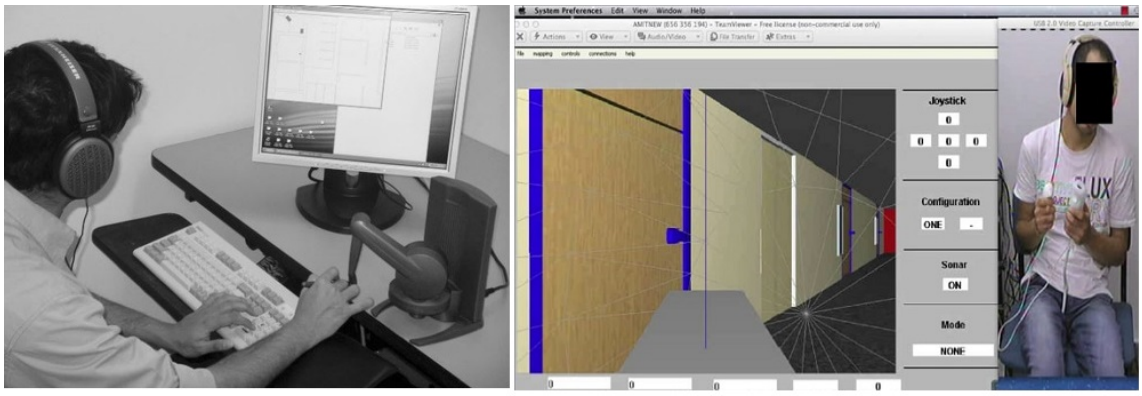
In contrast to vision, audio and haptic are undependable [19, 20, 21] - the information from cues can be misinterpreted and there are restrictions on both modalities – in audio, not all significant content can be transmitted through sound [19] and in the case of haptic, there is a need to be close to the objects [19, 20, 21]. Therefore, there is more benefit in using more senses (multimodality feedback), than using them individually. This fact makes devices very important in VEs since they are the main components in obtaining feedback. In other words, for a blind person to have good communication and obtain knowledge with the virtual environment is necessary to think about the interaction - the devices and the feedback they provide.

Traditional Interactions; In Human-Computer Interaction (HCI) **traditional interaction devices** has been the foundation of how we interact with technology, where the interactive system is the desktop computer and a fixed set of devices [22], e.g. screen, controller (or gamepad), mouse, keyboard. Despite heavy dependency on visual feedback, research efforts for accessible virtual environments for blind people have been made.

Digital gaming is one of the most active research areas trying to improve blind people's experience in virtual environments (e.g., [11, 9, 23]). *NavStick* [11], for instance, allows blind gamers to probe their surroundings by scanning a specific direction at a time with the gamepad thumbstick. And, in the following work [23] new insights regarding Spatial Awareness Tools (e.g. smartphone map, directional scanner) for visually impaired people are shared, such as preferences regarding position, orientation, and customization, among others. *Gonçalves et al.*, [9] explored how expert blind gamers played mainstream visual-centric games, as well as the strategies employed to successfully navigate the environment. Across these works, the importance of sound design is highlighted, as well as the ability to understand the surroundings to be able to navigate therein.

Another line of research using traditional interactions is the design of virtual environments to **support mobility training and/or gaining spatial knowledge of real-world locations** [24, 17]. For instance, *BlindAid* [24] (Figure 2.1 a)) is one of the virtual environment systems developed to help blind people with pre-planning exploration and cognitive mapping of unfamiliar environments. It uses multimodal feedback (haptic and audio), where the haptic hardware (Phantom) allows the person to move the avatar in the virtual environment and gives feedback close to a physical white cane, while the headphones provide sound cues when the avatar gets in contact with an obstacle. *Lahav et al.* explored a more accessible approach, with the use of Nintendo's Wii Technology [17]. This study also used multimodal feedback (haptic and audio), where the Wii remote (Figure 2.1 b)) functioned as the white cane to scan the surroundings, providing sound

cues from the speaker and vibrations, while the Nunchuck was used as the direction input device enabling movement in the environment. The inputs from the remote could be customizable where the blind person could configure the keybindings from the keyboard and mouse to the Wii remote physical buttons.



a) b)

Figure 2.1: Example of traditional interaction devices: a) User using the *BlindAid* system from [24]; and b) Blind person using the Wii-based system from [17].

Natural Interactions; In contrast to traditional interactions, **natural interaction modalities** mark a significant difference in HCI, particularly for blind and visually impaired people. The main difference with traditional interaction is that users are allowed to interact with the virtual environment as they are used to interacting with the real world in everyday life. It allows alternatives such as **speech** (e.g. voice commands), **gestures** (e.g. swipe), or **full body movement**. Gestures are the most commonly used interaction in this method, due to being one of the most important communication approaches, since can effectively express user demands [22]. For example, in the game *Alice Mystery Garden*[1], the player needs to change perspectives (1st or 3rd person) to overcome adversity, wherein the third person, gestures to push, pull, grab and rotate, are the same as it would be in the physical world. Another example is the gestures when using a smartphone (tap, swipe, rotate and tilt the smartphone) [25, 26]. Gestures can also be done with different devices - *LI et al.* [22] classify device inputs (Figure 2.2) into three categories: **wearable-sensor device-based input** - like the name implies, are devices that can be worn either fixed on the body or near the body in different positions, for example, VR Gloves (e.g. Gloveone), smartphones (e.g. PocketStrafe app[2]) or even VR default hardware (e.g. Head-Mounted Display (HMD) and VR controllers); **touch device-based input**, for example, stylus pen (e.g. Microsoft Surface Pen[3]); and **computer vision interaction device-based input** - which do not require to wear or maneuver external devices, where cameras are the main device, allowing hand tracking (e.g. Leap Motion

[1]Alice Mystery Garden - `https://vrgamecritic.com/game/alice-mystery-garden` (Last visited on February 27th, 2023).

[2]PocketStrafe - `https://pocketstrafe.com/` (Last visited on February 28th, 2023).

[3]Surface Pen - `https://www.microsoft.com/en-us/d/surface-pen/92fp8q09qhxc` (Last visited on February 28th 2023).

Controller[4], or Meta Quest 2 HMD[5]), body tracking (e.g. Microsoft Kinect) or eye-tracking (e.g. Tobii PCEye[6]).



Figure 2.2: Examples of the device input categories - a) PocketStrafe from [2], b) Microsoft Surface Pen from [3], c) Leap motion controller from [4]

For instance, *Steins et al.* show gesture-based imaginary devices [27], which implement two of the inputs mentioned previously - wearable (Essential Reality P5, wired glove) and computer vision (ten-camera OptiTrack system), allowing users to choose the input modality best suited for the task at hand by mimicking the use of physical input devices (Figure 2.3).

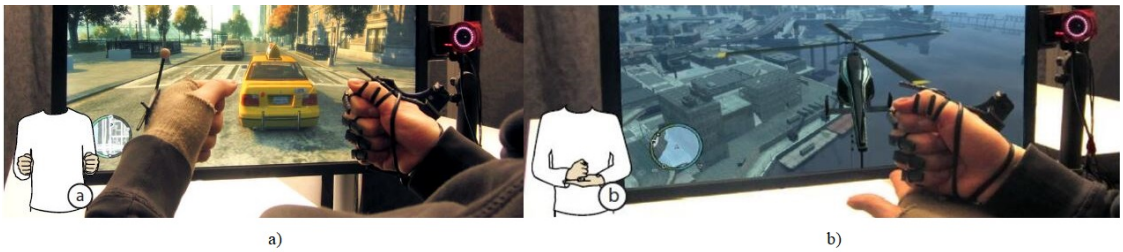


Figure 2.3: Examples of imaginary devices, mimicking input devices - a) Steering wheel for driving, b) Joystick for flying, both from [27]

In daily life, people do not just use gestures to communicate but also use voice (and others, e.g. eyes) to convey information [22]. Voice interactions refer to the communication between humans and machines through the use of spoken language [28]. And this technology became increasingly popular, with the rise of virtual assistants such as Siri, Alexa, and Google Assistant [29]. Speech is an interaction modality that can expand accessibility in HCI. One of the main advantages is enabling users to engage with the virtual environment without the need for controllers or input devices, allowing people with motor limitations to interact with the virtual environment without handheld devices (e.g. might require fine skills) or using gestures (e.g. requires upper body movement), offering a more natural and intuitive way to interact with virtual environments. Therefore is not surprising that voice interactions are an appealing subject for various fields (e.g. cultural

[4]Leap Motion - `https://www.ultraleap.com/product/leap-motion-controller` (Last visited on February 28th, 2023).

[5]Quest 2 - `https://www.meta.com/help/quest/articles/headsets-and-accessories/controllers-and-hand-tracking/hand-tracking-quest-2` (Last visited on February 28th, 2023).

[6]PCEye - `https://www.tobiidynavox.com/pages/pceye` (Last visited on February 28th, 2023).

institutions [30], gaming [31], or healthcare [32]). However, designing effective voice interactions can be challenging due to the complexity of the technology and the need to balance the user's sense of presence with the usability of the interface [28].

Overall, natural interactions go beyond practicality. Qualities such as pleasure, engagement, and immersion impact users more when using natural interactions compared to traditional ones. *Maurice et al.* in [33] compared gestures with two traditional interaction devices, both in terms of their pragmatic and hedonic qualities in two experiments: mouse vs gestures; and controller (Wii) vs gestures. The study shows that gestures in both experiments topped in hedonic and fun, while the traditional devices got higher scores in pragmatic and performance. Such qualities (e.g. immersion) can also be observed when using the real world. *Regal et al.* [34] developed a "scavenger hunt" which can be played either in the real world, where the setup is on the school building, using a smartphone as the assistive technology, **or**, played virtually with a 3D model of the school using audio as feedback on a desktop. According to the authors, the preference of the players to play this game is in using the real world. In the results of the digital survey provided by the authors, ten out of twelve thought the location-based game (mobile "version") was more fun. The reason is because of the full body movement where players quote - "*going around and being on the move*" [34], providing an immersion element throughout the game, where the player feels more involved.

Nonetheless, even with more active interactions, audio, and haptic feedback continue to show significance in virtual environments. For instance, in more immersive VR – e.g., using a Head-Mounted Display (HMD) – prior work has also shown the importance of audio and haptic feedback for understanding virtual environments [35]. Recent efforts include exploring alternatives for augmenting virtual objects with audio representations using a VR boxing use case [36], and supporting social VR experiences by dividing the social space into different social bubbles while conveying audio feedback accordingly [37].

The abovementioned research has contributed valuable knowledge to design more accessible virtual environments. For systems with desktop and smartphone implementations, navigation often relies on the keyboard or joystick (e.g., [38]), or the smartphone itself (e.g., [25]). Still, current technology now supports more immersive alternatives that blind people may leverage.

2.3 Locomotion in Virtual Reality

In the last several years, VR has gained significant attention with increasing availability and affordability, offering new possibilities for immersive experiences. Locomotion is an essential aspect of VR design and is vital to the overall User eXperience (UX) [1]. It enables users to engage with virtual environments in previously impossible ways, such as physically walking, running, or jumping.

2.3.1 Taxonomies

Prior research has extensively studied the numerous locomotion techniques proposed and implemented in commercial and academic contexts. Researchers actively try to define existing locomotion techniques and study essential elements such as immersion, ease of use, tiredness, and cybersickness, among others [1, 2, 3, 39, 40]). These elements play a key role in shaping the user's point of view of the technique, a crucial aspect for fostering a positive reception of the VR application [41]. Therefore, the number of new LTs has increased significantly over the years, making the use of locomotion taxonomies increasingly important in the field of VR research [3, 12]. These taxonomies help to establish a shared vocabulary and understanding of different LTs, enabling researchers to describe their experiments and findings more clearly and effectively [39, 41]. Moreover, the use of locomotion taxonomies can enhance the comparability of studies and publications, facilitating the meta-analysis of research results [39, 41].

Literature reviews often assign locomotion techniques into specific categories, creating taxonomies to group similar locomotion techniques together. To ensure that our overview of LTs is up-to-date (time of this writing), we will use a recent systematic review by *Martinez et al.* [3], which relies on known taxonomies [39, 42, 43] to categorize locomotion techniques and understand their exploration in the literature. The taxonomy classifies LTs into - Walking-based; Steering-based; Selection-based; Manipulation-based and Automated locomotion [3].

2.3.2 Locomotion Techniques

Walking-based locomotion. Techniques that rely on physical movements and actions [3]. Walking-based techniques can be further divided into three types: ***Full-gait*** - techniques that utilize the entire gait cycle, meaning - the time the foot is on the ground, and the time the foot is on the air [3, 44]. For example, *Real Walking* is a technique that implements a ratio of 1:1 of the mapped virtual environment and real-world (e.g. walking in a mapped parking lot [7]), where the user needs to fully walk throughout the space [3]; ***Partial-gait*** - techniques that do not require a full gait cycle [3]. For instance, in *Walking-in-Place* techniques, users solely move their feet while remaining stationary, causing movement in the virtual environment [8] [3, 45]; ***Gait-negation*** - techniques that use specific devices (Figure 2.4), that allow for a complete gait cycle experience while maintaining the user's stationary position [3] (e.g. [9], [10], and [11]). For example, *Omnidirectional Treadmill* cancel out the user's physical steps in any direction, enabling unrestricted virtual movement without physical displacement (e.g. Infinadeck Multidirectional VR Treadmill [12] - Figure 2.4 c)) [3].

Steering-based locomotion. Techniques where users may continuously specify a direction of travel achieved through various methods (e.g. eyes, or hands) [3, 46]. Steering-based can be

[7] VoodooDE mapping example - `https://youtu.be/JH6uWIwJZ40` (Last visited on March 9th, 2023).
[8] KAT loco - `https://youtu.be/SvTbB19bvIw` (Last visited on March 9th, 2023).
[9] MSE Weibull device - `https://youtu.be/nQR49JGySTM` (Last visited on March 9th, 2023).
[10] Unidirectional treadmill - `https://youtu.be/MWNeTNcMIFM` (Last visited on March 9th, 2023).
[11] CirculaFloor - `https://youtu.be/rYsvB2y2Ero` (Last visited on March 9th, 2023).
[12] Infinadeck treadmill - `https://www.infinadeck.com/landing/` (Last visited on March 9th, 2023).

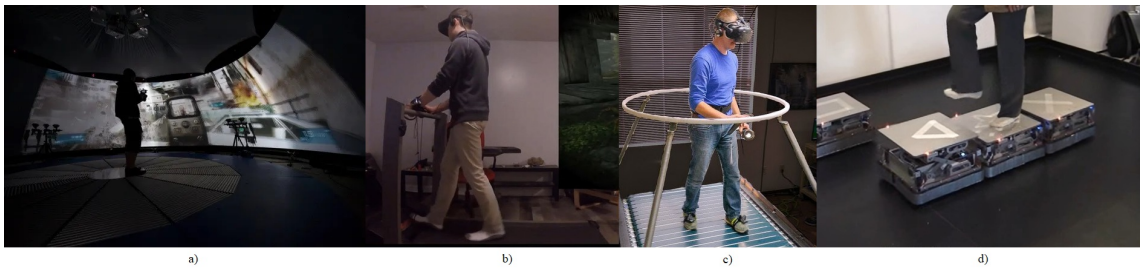


Figure 2.4: Examples of Gait-Negation LTs - a) Low-Friction Surface from [9], b) Unidirectional Treadmill from [10], c) Omnidirectional Treadmill from [12], and d) Step-based Device from [11]

divided into two types: ***Spatial steering*** - locomotion techniques that allow manipulation of a device to control the direction, usually involving the use of another input to enable the technique (e.g. pressing a button). [3]. For example, *Head-directed Steering*, or *Gaze-directed Steering* techniques move the user in the direction they are looking by utilizing the HMD (e.g. VR Dash[13]) [3, 46]. ***Non-spatial steering*** - techniques constrained to particular situations because rely on specialized devices that vary depending on the type of steering required [3, 46]. For instance, in *Joystick-directed Steering* users can control their virtual movement by manipulating a joystick to control their forward and backward motion as well as their heading within the virtual environment (e.g. Flight Stick[14]) [3, 47].

Selection-based locomotion. Techniques that allow users to select a destination or path to travel [3]. For example, *Point & Walk* techniques enable users to select a destination (by pointing), but instead of instantly transporting them to that location, they have to walk through a portal to reach their selected destination (e.g. projected portal[15]) [3].

Manipulation-based locomotion. Techniques where users may manually control their position in the environment, such as dragging to a particular position [3]. For example, *World-in-Miniature* techniques enable users to change their location in the virtual environment by manipulating a miniature version of themselves within the environment [3] (e.g. Brass Tactics [16]).

Automated Locomotion. Involves the system controlling the user's movement [3]. For instance, *Automated Steering* techniques resemble traditional steering methods, but with the system determining both the direction and when movement occurs (e.g. VR Roller Coaster[17]) [3];

Locomotion Techniques Overview. Ultimately, *Martinez et al.* [3] found that Walking-based Locomotion is the most prevalent, followed by Steering-based and Selection-based. Less common techniques fall under Manipulation-based and Automated locomotion.

Similarly, in prior work, Boletsis [40] found that most VR locomotion techniques rely on physical interaction, harnessing physical motion cues to navigate within VR environments. Still, reviews and existing comparisons among a subset of techniques showcase the pros and cons of each technique, sometimes with contradicting findings. For instance, joystick-based techniques are

[13]VR Dash - `https://youtu.be/txaYe0y9ov4` (Last visited on March 9th, 2023).
[14]Protas VR Flight Stick - `https://youtu.be/ucI0zGf19FQ` (Last visited on March 9th, 2023).
[15]Point & Walk - `https://youtu.be/mv7LjX_ggj0` (Last visited on March 9th, 2023).
[16]Brass Tactics - `https://youtu.be/f2GCI75Velw` (Last visited on March 9th, 2023).
[17]VR Roller Coaster - `https://youtu.be/Ei0fgLfJ6Tk` (Last visited on March 9th, 2023).

sometimes found to outperform the others [2], but specific comparisons often find more efficient techniques (e.g., *Point & Teleport* [48], or *Redirected Walking* [49]). These differences support the need to investigate prominent techniques under different scenarios. Still, the accessibility of locomotion techniques is not addressed in prior comparisons or literature reviews, despite the known challenges experienced by people with disabilities with VR technologies [4].

An exception is the *Locomotion Vault* [1], a database and visualization consolidating information on 109 locomotion techniques. It contains a selection of 19 attributes derived from the analysis of the literature and two attributes proposed by the authors - one of which is accessibility ("*how easy is adopting this LT for users with disabilities?*" [1]). Despite the step forward, accessibility is defined in terms of the motor ability required to use the techniques, which is likely due to prior research on VR for people with motor impairments (e.g., [5, 50]). Nevertheless, there is very limited knowledge of the accessibility of locomotion techniques for people who are blind.

2.4 VR Locomotion for Blind People

Visual feedback is the primary means of interaction in current VR experiences, posing major challenges for blind people to both understand and navigate the environment. Fortunately, VR systems besides including other sensory feedbacks, such as audio and haptic [51], also allow different interactions - traditional (e.g. thumbstick) and above all natural interaction (e.g. gestures, speech).

Locomotion techniques aiming to create immersion often draw inspiration from real-world settings, translating physical movements from real into virtual environments. An existing approach is to walk in the real world, producing the same effect in the virtual world. This has been used in past work on O&M training [6]. *Thevin et al.* [6] designed an affordable and accessible VR system for O&M classes, using a smartphone and a Bespoke headmount (Figure 2.5 a)). The system provides multimodal feedback (visual and audio) and permits users to move in both environments on a 1:1 scale (real and virtual) without extra equipment, where the VE illustrative scenarios are crossing a street in a pedestrian crossing, and a traffic light (Figure 2.5 b) and c), respectively).



a) b) c)

Figure 2.5: *X-Road* prototype and configuration - a) X-Road prototype consists of a headset, smartphone, and 3D-printed holder, b) Overview of the VE configuration representing a street with a pedestrian crossing and c) Overview of the VE configuration representing a traffic light crossing, all from [6]

However, the equivalence to real walking imposes constraints as either the experience has to

be designed for small spaces or a large space for using the device is needed. A way to deal with physical space restrictions is to use walking-based techniques where users stand in place. Prior explorations with blind people include either a treadmill or trackers to detect steps [51, 52], which require specialized equipment. *Kreimeier et al.* [51], compare four different implementations of VR locomotion (Figure 2.6) – two treadmills, trackers on the ankles, and a joystick, to answer "*if and how which type of locomotion in VR is suitable and interesting for blind and visually impaired people*" [51]. The authors found the latter (joystick) to feel safer, more precise, and intuitive. In addition, participants felt walking on treadmills was unnatural. Still, the evaluation was based on the users' subjective perceptions alone.



Figure 2.6: *BlindWalkVR* locomotion implementations - **Treadmill Locomotion**: a) The Cyberith Virtualizer, b) Overshoes and optical sensors integrated into the ground plate; c) Virtuix Omni, d) Inertial Measurement Unit sensors on the special shoes; e) **Ankle locomotion**, f) mounted VR trackers; and **Joystick locomotion**: g) Acer AH100 Windows Mixed Reality HMD and a Logitech G933 headset, h) left controller's joystick, all from [51]

Similarly, *Han et al.* [52], conducted a study utilizing a VR omnidirectional treadmill and VR trackers (attached to ankles) to examine the impact of walk-in-place and physical walking on spatial information acquisition related to paths and obstacles. The study results suggest that the participants demonstrated better retention of route spatial information when trained with the treadmill while remembering better the obstacles through training with the trackers. Moreover, the authors state that using trackers might be better/more suitable for wayfinding training when exploring obstacles compared to omnidirectional treadmills.

Other techniques, exclusively designed for blind people, use their mobility skills and navigation aids to support virtual navigation. In particular, prior work has proposed using a white cane, augmented with haptic, force-feedback to simulate obstacles and real-world navigation [8, 53, 54].

For example, *Siu et al.* [8], examine a novel white cane controller (Figure 2.7 a)). The controller provides human-scale force feedback using a novel braking system alongside surface texture associated with tapping and sweeping. The cane provides audio cues based on the geometric and material properties of the objects in the virtual environment. However, in the results, participants had split opinions regarding the familiarity of the sound effect of the cane. This was caused by participants' preferences regarding the material of the tip (plastic or metal), since in the study the sound and haptic feedback were of a metal-tipped cane.

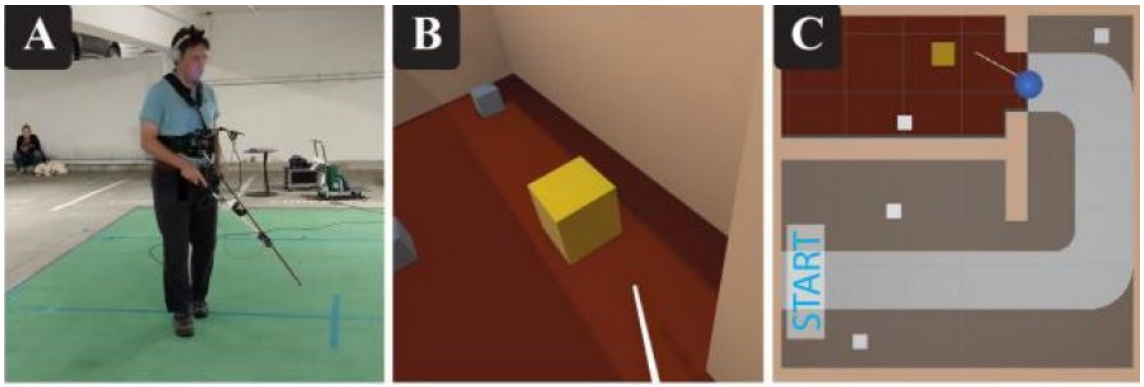


Figure 2.7: Exploration with novel controller - a) Participant navigating with the controller, b) 1st person view of the VE, and c) Overview of the VE map, all from [8]

Alternatively, prior work has also explored teleporting to specific locations selected from lists of objects, people, or points of interest [55]. Such approaches significantly ease reaching specific destinations but at the cost of freedom in moving through the whole environment.

While these novel, specialized approaches provide valuable contributions to the field, they are generally unavailable in current VR applications – for instance, due to the need for additional equipment, and/or because the prototypes are used in research contexts alone. In addition, designers often carefully choose the locomotion method as it affects the experience [56, 57], meaning specialized techniques could be complex (or sometimes unfit) to integrate into existing experiences – e.g., walking with a white cane significantly differs from a teleporting technique. For these reasons, VR applications are currently deeply tied to the locomotion techniques selected by their designers. Understanding the potential of current prevalent locomotion techniques to support accessible experiences may increase and diversify blind people's access to (mainstream) VR experiences.

2.5 Discussion

Natural interactions compared to traditional ones are still in their infancy, but bring new ways of interacting with VEs, creating more inclusive, accessible, and immersive environments. However, it does not mean that natural interactions are an omnipotent solution, where we see traditional interactions still being used, and the abovementioned research proves that traditional interactions

have qualities that depending on the system are more beneficial compared to a more instinctive interaction. Nonetheless, the hedonic qualities that natural interactions bring, are relevant to give importance to studying continuation and expansion to other research fields (e.g. gaming).

Through technology development, archived ideas using natural interactions are reappearing giving VR research new challenges. One is the large number of LTs, where frameworks for categorizing the different LTs have been studied to give a more general perspective of what already exists in the literature [1, 2, 3, 39, 40]. However, even with numerous LTs, accessibility to people with sensorial limitations is still a huge challenge, where there is more focus on motor limitations since VR locomotion is generally viewed with body movement [5]. Since VR applications rely on visual information, blind and visually impaired people are neglected due to barriers in navigation and/or orientation in VEs, emphasizing the importance of this study or general research line to blind people.

With the abovementioned research on blind VR locomotion, it can be said that there might not be a definitive or foolproof LT, since user descriptions are counted on the next study decisions, entering the user's preferences domain. With possibly infinite variations of user preferences, is a challenge to provide "the best UX". *BlindWalkVR* [51] is a good example, where the authors explore suitable locomotion implementations for blind and visually impaired people, where natural interactions theoretically would expected to be the preferred interaction. However, traditional interactions got higher scores from the participants. One possible reason is the familiarity with current devices, such as the joystick (e.g. video gaming), among others, causing it to be faster and more intuitive, since it's more practical than moving the full body in a manner that people are not used to, making it like participants described in *BlindWalkVR* as "*unnatural*" [51]. Moreover, in *VR Wayfinding* [52] there is an indication that different techniques (*Walking-in-Place* and *Real Walking*) based on the same type of locomotion (Walking-based) give different spatial information. However, the prior research in accessible VR for blind people represents a small array of available locomotion techniques. In that regard, there is little to no knowledge of how different locomotion techniques may support accessible VR experiences for blind people and how VR elements affect them (e.g. levels of comfort, and ease of use).

Overall, prior research has made significant contributions, frequently relying on custom-made solutions that demand specialized equipment or extensive/unrestrictive spaces. Unfortunately, this specificity often restricts the widespread adoption and implementation of such techniques due to their intricate setup requirements.

This related work highlights the importance of considering both traditional and natural interactions in VEs, as well as designing accessible and effective locomotion systems with a focus on blind people.

Chapter 3

Cataloging the Accessibility of VR Locomotion Techniques

In light of the insufficient attention given to accessibility in VR systems, researchers recommend taking action to address the challenges (e.g. [4, 58, 59]). One (challenge) is mobility in immersive virtual environments for blind and visually impaired people. To make locomotion in VEs accessible, it is imperative to investigate different LTs and their potential effectiveness. However, with a vast array of LTs proposed in the literature, and despite the known challenges experienced by people with disabilities with VR technologies, the accessibility of locomotion techniques is not addressed (e.g. [2, 49]). An exception in the literature is the ***Locomotion Vault*** [1], a comprehensive database of LTs sourced from scientific papers, gaming venues, game house websites, social media, and blog posts. However, despite the step forward, **accessibility is defined in terms of the motor ability required to use the techniques**.

Nonetheless, the database currently catalogues 109 LTs with the implementation of UX attributes proposed in the literature "*to describe the function and properties of the LTs so that they can be discriminated based on how they work*" [1]. The attributes (21 in total) were obtained from academic literature apart from two that were proposed by the authors - video gamer and accessibility.

Our goal was to gain more knowledge about VR locomotion techniques accessibility, which would help us select techniques to further investigate in more detail. Therefore, with a significant amount of VR locomotion techniques (109 LTs), and since the database regarding accessibility focuses on motor ability, we concluded that the accessibility attribute within the database can and should be explored more in-depth.

3.1 Categories for Accessibility

Designer guidelines help with the design of undesirable features and aspects of a virtual environment, avoiding barriers and including people with disabilities in obtaining a more desirable end experience [60]. We investigated two accessibility guidelines when using VR, AR or Mixed Reality (XR) technologies and traditional video gaming, such as World Wide Web Consortium (W3C)

XR Accessibility User Requirements[1] and *Game Accessibility Guidelines*[2], to select the baseline categories for our analysis.

To select the categories from the guidelines, we first considered accessibility challenges relevant to locomotion techniques (e.g. inputs, disabilities), while in parallel we compared with the UX attributes (Table 3.1) in the database to maintain relevance.

Table 3.1: Complete description of the UX attributes proposed on the locomotion vault [1]

UX Attribute	**Type**	**Values**
Posture	Categorical	Any; Standing; Seated
Granularity	Ordinal	Low; Medium; High
Speed	Ordinal	N/A; Low; Medium; High
Large VE	Ordinal	N/A; Low; Medium; High
Multitasking	Ordinal	N/A; Low; Medium; High
Energy	Ordinal	Low; Medium; High
Spatial Awareness	Ordinal	Low; Medium; High
Vection	Ordinal	Low; Medium; High
Nausea	Ordinal	Low; Medium; High
Embodiment	Ordinal	Low; Medium; High
Intuitiveness	Ordinal	Low; Medium; High
Accessibility	Ordinal	Low; Medium; High

Descriptions
What posture(s) does the locomotion method support?
How precise can the control of the location in VR be with this LT?
How fast does the user travel in the VR space?
Does the LT allow movement in large VR environments?
To what extent can the user perform other tasks at the same time as locomotion?
How much energy does the user spend to move in VR?
Does the LT allow the user to be aware of their surroundings [...] in the VR environment?
How much do the users feel that they are moving?
How much simulation sickness can this LT induce in a novice user?
How much does the LT invoke the sense of having a body in VR?
How easy is it to adopt this LT without any training or learning?
How easy is adopting this LT for users with disabilities?

Next, the author conducted informal exercises to analyse some techniques and identify any missing information, sharing the findings and insights with the team. Following this, two additional researchers collaborated in a similar process to increase validity, reliability, discussion, and perspectives, obtaining the proposed categories (Table 3.2) in the following section.

[1]W3C XR Accessibility - `https://www.w3.org/TR/xaur/` (Last visited on March 16th, 2023).

[2]Game Accessibility Guidelines `https://gameaccessibilityguidelines.com/` (Last visited on March 16th, 2023).

Table 3.2: Baseline categories and their origins/sources, and the categories proposed to augment the locomotion vault [1]. For a complete description of the proposed categories see Table 3.3

Baseline Categories	O'Connor et al. [1]	Ellis et. al [2]	Di Luca et. al [1]	**Proposed Categories**
Alternative Input Modalities	X			Multimodality
				Input Modality(ies)
				Input Type(s)
				Output Modality(ies)
				Output Type(s)
				Alternative Inputs
Time Sensitive		X		Time Constraints
Accessibility (Motor)			X	Upper Limbs
				Fine Skills
				Lower Limbs
Posture			X	Play Area
Cognitive	X	X		~~Cognitive~~
Speech	X	X		Speech
Hearing	X	X		Hearing
Vision	X	X		Vision

3.2 Categories

In this section, we describe the 14 proposed categories and their respective descriptions. The proposed categories are collated from two accessibility guidelines and the database (Table 3.2). However, we made adjustments to the categories, considering that the scope of the sources included broad or specific scenarios with different technologies (other than VR). In addition, there was also the need to verify the categories already incorporated into the database, to keep the catalogue relevant.

Multimodality. With multiple and possible combinations of modalities to support various user needs (see subsection 2.2), came the necessity to identify if the LTs were multimodal or unimodal (in the context of virtual environment interactions). For example, if a technique uses just gestures or a combination of two or more modalities. Therefore, to determine whether the technique combines multiple, or single, input modalities, we added ***Multimodality***, with the following description - *Does the locomotion technique have more than one input modality?*.

Modalities. Nonetheless, understanding and identifying which modality(ies) are being used helps analyse which LT supports accessibility-related user requirements, providing valuable insights into the technique's design. Thus, adding the category ***Modalities***. However, users can interact with the virtual environment (input) and receive feedback (output), where the choice of input and output modalities can significantly affect the user experience in VR. Therefore, we di-

vided ***Modalities*** in ***Input*** and ***Output***, identifying which movements or commands the user is providing to the VR system and what feedback is the VR system providing to the user. Considering the multiple interactions between the user and the VR system, was necessary to distinguish the ***Modality(ies)*** (e.g. gestures) and the ***Type(s)*** (e.g. hand tracking) being used. Additionally, the rule of inclusion is the same as in *Locomotion Vault* [1], where a new modality and type were added if there was none already created that could not describe it. Therefore, ***Modalities*** is divided in four categories - ***Input Modality(ies)***, ***Input Type(s)***, ***Output Modality(ies)*** and ***Output Type(s)***, where we want to know - *What are the **input** and **output** modalities and types used on the locomotion technique?*.

Alternative Inputs. In addition, we identified techniques that provided different inputs with the same modality to provide the exact movement to the VR system (e.g. moving one's head back and forth could create movement, as could jogging in place, which causes the head to bob). There is an importance to identify and differentiate from the ***Modalities*** because if the alternative exists, implies a more robust and accessible locomotion technique, which increments the number of users that can use the technique - adding the category ***Alternative Inputs***, where we identify if an alternative exists and what are the alternatives (e.g. leg movement) which would provide the same movement - *Does the locomotion technique offer different inputs for the same input requirement? Which?*.

Time Constraints. VR locomotion techniques can have time limits depending on the implementation and the experience being created. These constraints could be to balance the technique - for instance, in some teleportation techniques, designers implement "cooldowns" or limitations to prevent frequent teleportations (e.g. [61]), or to reduce disorientation, such as implementing vignetting or gradually fading to black in the technique (e.g. [62]). While, the previously mentioned examples are the general/typical implementations of time limits in an LT, making sure the user is not required to perform multiple inputs at the same time and providing enough time for the user to read, react, listen, and process information (e.g., [1, 2]) should also be considered to target a wide range of users, introducing ***Time Constraints*** - *Does the locomotion technique demand a time limit?*.

Impairments. Finally, based on the abovementioned guidelines, there are at least five general types of disabilities: ***Hearing/Auditory***, ***Vision/Visual***, ***Speech***, ***Motor/Physical***, and ***Cognitive***, which a user may have one or a combination of these disabilities. Understanding that gestures or general body motions are the main focal points of the immersive quality in VR, is unavoidable giving a more detailed evaluation of the motor impairment in comparison with the other general types. Also, this was possible since the analysis was with visual content, where user movement can be seen (see section 3.3). The motor impairment was divided into four categories:

- ***Upper Limbs*** - *Does the LT require upper limb motion (e.g. head, arms, trunk, wrists)?*
- ***Fine Skills*** - *Is finger coordination required?*
- ***Lower Limbs*** - *Does the LT require lower limb movement (e.g. jumping, walking)?*

- ***Play Area*** *- What play area does the LT support (e.g. standing, seated, both)?*

Table 3.3: The proposed categories and descriptions

Category	**Description**
Multimodality	Does the LT have more than one input modality?
Input Modality(ies)	What are the input modalities used on the LT?
Input Type(s)	What are the input types used on the LT?
Output Modality(ies)	What are the output modalities used on the LT?
Output Type(s)	What are the output types used on the LT?
Alt. Inputs	Does the LT offer different inputs for the same input requirement? Which?
Time Constraints	Does the LT demand a time limit?
Upper Limbs	Does the LT require upper limb motion (e.g. head, arms, trunk, wrists)?
Fine Skills	Is finger coordination required?
Lower Limbs	Does the LT requires lower limb movement (e.g. jumping, walking)?
Play Area	What play area does the LT support (e.g. standing, seated, both)?
Speech	Is voice projection necessary for effective use of the LT?
Hearing	Is audio necessary for effective use of the LT?
Vision	Is vision necessary for effective use of the LT?

As mentioned in the section 2.2, gestures are the most commonly used interaction (natural interaction modalities), and with device inputs is possible to target specific body parts or the full body. This fact made the ***Upper*** and ***Lower Limbs*** categories a must-include since target the majority of the LTs. However, VR systems also allow stationary interactions (e.g. avoiding being constrained by the physical world), where the user can interact with the system with familiar methods (e.g. thumbstick), being necessary to include ***Fine Skills*** and ***Play Area***.

Typically, to augment the *Locomotion Vault*[1], it's redundant to include a category (***Play Area***) equal/similar to one already introduced ("***Posture***" category/attribute on the *Locomotion Vault* [1]). However, when analysing the LTs in the initial phase, there were biased answers on ***Lower Limbs*** because there are no specific notes about the relation with posture and the use of the lower limbs on the *Locomotion Vault* [1]. Not all standing techniques require lower limb movement, and not all seated techniques are indifferent to lower limb movement - meaning that standing or seated, do not validate entirely the usage and non-usage for the lower limbs (e.g. when using a VirZoom[3] bike, the user is seated but is using the legs). With a wide range of limitations, analysing with the "***Posture***" would incorrectly imply that certain users could not use the locomotion technique. Therefore, we included the category ***Play Area*** to differentiate from the previous ("***Posture***") and also pass a more concise message of the physical space that surrounds the user, where the ***Lower Limbs*** describe if there is movement even in a stationary position in a limited space.

[3]VirZoom - `https://youtu.be/x_X4SsrWoQ0` (Last visited on March 9th, 2023).

The importance of mobility as a skill for VR has rendered the platform inaccessible to certain users. Fortunately, gestures are not the only natural interaction that can provide virtual movement with current technology (see section 2.2) - voice and eye interactions can cover mobility limitations, where guidelines such as *XR Accessibility User Requirements* suggest speech and eye-tracking as input modalities. However, with complex VEs processing information can be overwhelming even with alternative modalities. Therefore, the following subcategories were included not only due to guidelines but also due to the possible combinations of modalities in LTs.

- ***Speech*** *- Is voice projection necessary for effective use of the locomotion technique?*
- ***Hearing*** *- Is audio necessary for effective use of the locomotion technique?*
- ***Vision*** *- Is vision necessary for effective use of the locomotion technique?*

We initially attempted to analyse the need for cognitive abilities, but we encountered a significant degree of subjectivity that we felt would compromise the accuracy of the analysis. Consequently, we decided to exclude ***Cognitive*** from our analysis and instead focused solely on the remaining categories. The following Table 3.4 summarises the proposed categories with their typing and values. The metric system in our analysis is similar to the *Locomotion Vault* [1] - two categories are binary type (e.g. *Yes* or *No*), six are ordinal, with three possible values (e.g. *Required*, *Desirable*, *Indifferent* **or** *Standing*, *Seated*, *Both*) and the remaining are categorical.

Table 3.4: The typing of the proposed categories and the corresponding values

Categories	Type	Values
Multimodality	Binary	Yes; No
Input Modality(ies)	Categorical	None; Gestures; Hand Controller; Foot Controller
Input Type(s)	Categorical	[**Gestures**] - Head Tracking; Hand Tracking; Controller Tracking; Ankle Tracking; Body Tracking; Accelerometer; [**Hand Controller**] - Buttons; Thumbstick; TouchPad; Mouse; Keyboard; [**Foot Controller**] - Treadmill; Board; OmniTreadmills; Pedals; Robots
Output Modality(ies)	Categorical	Sense Perception
Output Type(s)	Categorical	Vision: Proprioception; Haptic; Audio
Alternative Inputs	Categorical	None; Head Movement; Arm Movement; Leg Mov.
Time Constraints	Binary	Yes; No
Upper Limbs	Ordinal	Required; Desirable; Indifferent
Fine Skills	Ordinal	Required; Desirable; Indifferent
Lower Limbs	Ordinal	Required; Desirable; Indifferent
Play Area	Categorical	Standing; Seated; Both
Speech	Ordinal	Required; Desirable; Indifferent
Hearing	Ordinal	Required; Desirable; Indifferent
Vision	Ordinal	Required; Desirable; Indifferent

3.3 Procedure to Catalog the Techniques

We analysed 107 LTs (two LTs were excluded) on *Locomotion Vault* [1], where two additional researchers were included to increase validity, reliability, discussion, and perspectives.

The analysis was based on content recorded in the database if available (e.g. description, images, animated images, links to multimedia content - mainly the YouTube video platform), and if not (available), the videos were searched manually with reference to the citations, creator's name, or by video games used as examples, also recorded on the *Locomotion Vault* [1]. For this reason, we approached with high-level categories, exposing general but concise information that the LTs provide when using them in an accessibility context. Furthermore, we constrained (C1-C3) the protocol to remove ambiguity in the analysis (Table 3.5).

The author and the two additional researchers conducted informal exercises to categorize 23 techniques and identify any missing information. In case the categories were not sufficient to analyse the technique, we would add the necessary categories and repeat the process, until arriving at a consensus. Finally, the remaining 86 LTs were analysed in pairs, where the two additional researchers divided the techniques between them and each paired with the author. Additionally, if there was a difference in one of the pair analyses, the remaining element would analyse the locomotion technique for further discussion until agreement.

Table 3.5: Protocol constraints

Constraints	**Application Example**
C1. Analyse only what the video provides, and if necessary - the scientific publication if it was based on one.	Do not assume other types of hardware that could be used, for example, accessibility possibilities (e.g. switches).
C2. ***Modality(ies)*** are only considered if they are directly related to the LT.	Head tracking is not assumed as a modality on the LT if the only purpose is camera control in the VE (e.g. analogue stick can be an alternative).
C3. ***Fine Skills*** require finger coordination or desirable if just holding VR controllers.	LT triggered by a button click is not counted as fine skills (e.g. possible to push the button with the stump of the hand).

Of all 109 LTs, three techniques could not be analysed the same way as the others. On the analysis, it was detected that the video supplied in the *Locomotion Vault* [1] for the "***Snap N' Motion***" technique did not correspond with the description recorded in the database. And, little information was found (e.g. [4]). The solution to analyse the technique was to solely rely on the description and the two UX attributes ("***Hardware***" and "***Motion Direction***") on the *Locomotion Vault* [1] since the video suggested on the database would compromise the analysis. Regarding the other two techniques, the entry of "***Walk in Place***" on the database was a general description of Walking-in-Place techniques (see subsection 2.3.2) rather than a specific technique that uses

[4]XinReality Snap N' Motion - `https://xinreality.com/wiki/Snap_N%27Motion` (Last visited on March 24th, 2023)

the Walking-in-Place method, this is reinforced when the video suggestion redirects to a YouTube content that speaks of various ways to walk in place. Additionally, the database has "***Similar Methods***" in each recorded technique to enable easy and faster search between the same types of techniques, and "***Disambiguation***" to describe the difference or what differs from the similar techniques. Therefore, based on the database information, "***Walk in Place***" has four similar techniques, and no entry about "***Disambiguation***". Due to subjectivity and redundancy, we excluded the "***Walk in Place***" technique from our analysis. Lastly, the analysis of the "***Turning + Dashing Teleport***" entry raised uncertainty due to inconsistencies between the visual content and the provided description in the database. Furthermore, the visual content appears to be a duplicate of another locomotion technique ("***Telepath***") but recorded in a gameplay scenario, casting doubt on the accuracy of the entry. As a result, the "***Turning + Dashing Teleport***" was excluded from the analysis.

3.4 Findings

In this section, we present our findings by descriptive analysis of the main five proposed accessibility categories - ***Modalities*** (including ***Multimodality***); ***Alternative Inputs***; ***Time Constraints*** and ***Impairments***.

3.4.1 Modalities

Despite evidential information that multimodality has more advantages (e.g. [22]), the analysis shows that there are more unimodal (63 LTs) than multimodal techniques (44 LTs) on the *Locomotion Vault* [1]. Nonetheless, ***Gestures*** is the mostly used input modality (see subsection 2.2) (Figure 3.1), where ***Controller Tracking*** is the most frequent input type used in VR techniques within the database (52 LTs), followed closely by ***Buttons*** (49 LTs) in the ***Hand Controller***, possibly due to the performance of traditional interactions [33], and the familiarity of input (e.g. gamepad in video gaming) [51]. It's worth noting, that even though ***Gestures*** is the most used modality, natural interactions such as voice commands were not found in the recorded techniques.

In contrast to input modalities, output had more multimodal LTs (82 LTs) than unimodal (25 LTs) (Figure 3.2). Vision influence in unimodal and multimodal domains is not a surprise (Figure 3.2), since there is a high dependency between VR and visual information. However, this distinction (with output types) is not only related to the study being based on the *Locomotion Vault* [1] but also to the restrictions made in the protocol. In this study, we did not include auditory feedback that comes from the virtual environment sound design because it's necessary for the locomotion technique to maintain its feedback regardless of the changes in the virtual environment that will be implemented. Additionally, the content used to analyse needed to showcase the information (LT feedback). Therefore, our analysis shows that only one locomotion technique integrates all the categorized output types (Figure 3.2).

Moreover, the two remaining output types (***Proprioception*** and ***Haptic*** - Figure 3.2), are related to the two input modalities analysed as ***None*** (Figure 3.1). The two LTs (***Galvanic Stimu-***

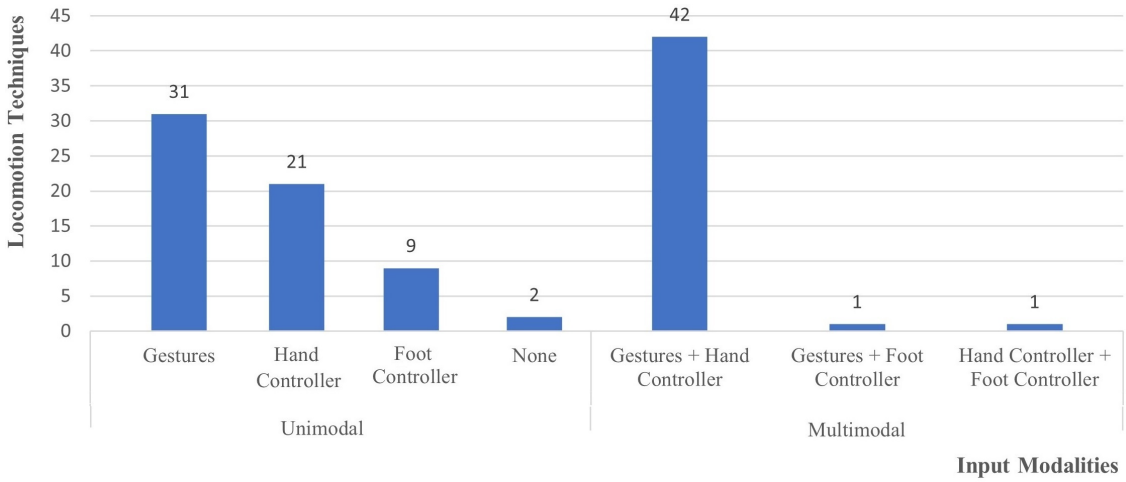


Figure 3.1: Input modality distribution - Unimodal and Multimodal

lation [5] and ***Haptic Feedback*** [6]) approach is to remove/decrease motion sickness by targeting the users' senses (e.g. stimulating the inner ear with electrical impulses), with specific devices where the techniques do not have an input modality that provides virtual movement in the environment.

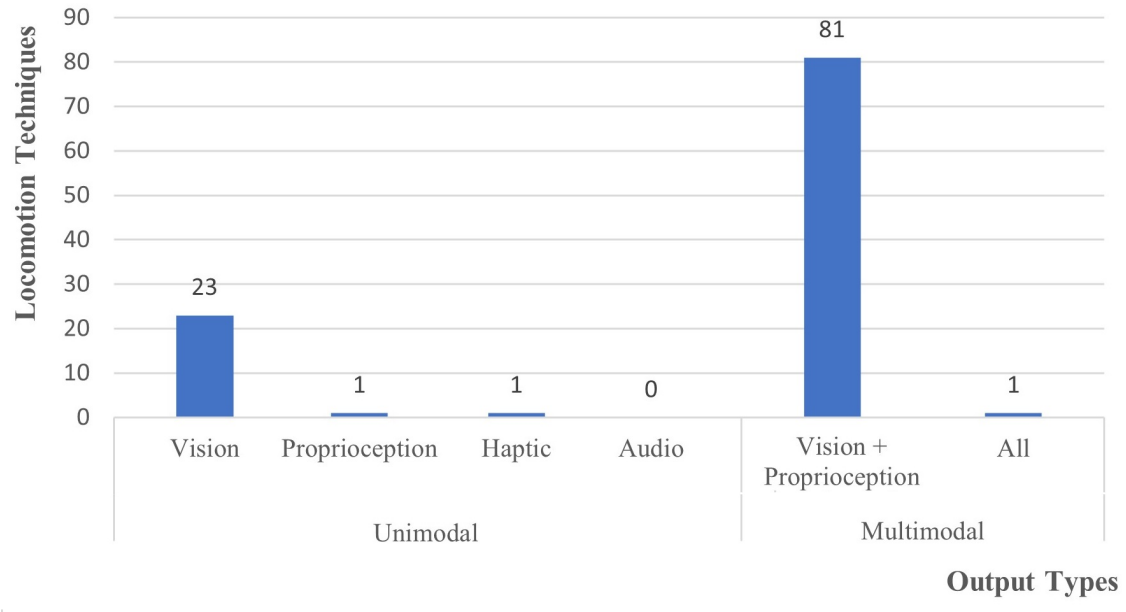


Figure 3.2: Output type distribution - Unimodal and Multimodal

3.4.2 Alternative Inputs and Time Constraints

We also found out that the recorded locomotion techniques had fewer alternative inputs with the same input requirements, which we identified only three LTs with alternatives. Moreover, 101 LTs did not demand a time limit to input a command. It does not necessarily mean that the technique

[5] Galvanic Vestibular Stimulation - `https://youtu.be/OlXYqfQHNuA` (last visited on March 24th, 2023).
[6] WalkingVibe - `https://youtu.be/vPCs90BlhzQ` (last visited on March 24th, 2023).

provides enough time for any limitation, but allowing "limitless" time (action will not happen until input), at least or to some extent, decreases the user getting overwhelmed by the environment.

3.4.3 Impairments

Finally, the data from the impairments (Figure 3.3 a)) tells that most of the LTs require ***Upper Limbs***, while the majority are indifferent about ***Lower Limb*** movement. This result might be related to the findings on the *BlindWalkVR* [51] regarding the walking techniques (treadmills and ankle trackers), where the users feel overwhelmed when walking (stationary or around) in the real world, which results in the community focusing more on welcoming methods to counter negative descriptions. Moreover, this non-evidential observation is complemented by traditional input modalities since complex finger coordination is beneficial to the majority of the LTs evaluated, for example, a simple button click translates into a complex action in the system, removing overall complexity, and decreasing the overwhelming feeling. In this context is not surprising that the ***Play Area*** has more LTs that can be used in both ways (***Standing*** and ***Seated***) (Figure 3.3 b)), the lower limb movement indifference is correlated to the amount of space the user has, allowing for preference gameplay, where the user decides which is more comfortable (if has a choice).

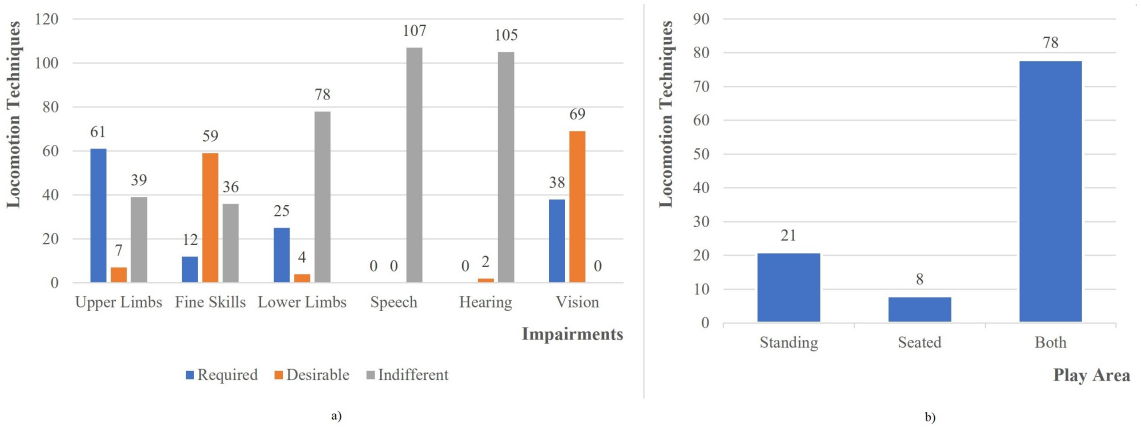


Figure 3.3: Impairments with corresponding values - a) [Required, Desirable, Indifferent]: Upper Limbs, Fine Skills, Lower Limbs, Speech, Hearing, Vision; and b) [Standing, Seated, Both]: Play Area.

It was previously noted that no LTs were identified with voice command inputs and audio outputs for locomotion. This outcome may be due to the applied constraints and focus of the study, which primarily centred around the use of multimedia content. So, it's no surprise the indifference in ***Speech*** and ***Hearing*** impairments (Figure 3.3 a)). However, opposite to the two previous categories, ***Vision*** does not have a single locomotion technique that is ***Indifferent***, and surprisingly it ended up as a ***Desirable*** rather than a required one. This result could be related to what is mentioned in section 2.2 about the differences between interactions, in the context, that by moving the body, proprioception provides some but definitely more information (compared to just being seated).

3.4.4 Summary

Overall, our findings in terms of accessibility show that VR locomotion techniques in the Locomotion Vault [1] vary for people with different disabilities (e.g. blindness, motor impairment). Blind people, for instance, are challenged by the visual content, however, multiple alternatives are possibly accessible to them, since the majority of the techniques use gestures as an input modality. In contrast, and generally speaking, people with motor impairments cannot use the majority of the techniques (depending on the impairment and technique availability), due to the focus on movement. Nonetheless, techniques with less physical demands, for example, alternatives focused on traditional interactions can be possible options. People with hearing and speech impairments have broader freedom in VR locomotion techniques. There is little to no technique that requires or is desirable to use audio or speech to make a virtual movement. Nonetheless, while hearing-impaired people are not as restricted as the blind and motor-impaired people, there could be limitations regarding the immersion aspect, since only two techniques in the database showcase haptic feedback integrated into the technique. Vibrations or haptic feedback can also be used to convey auditory feedback through touch, enhancing the experience [9].

3.5 Extending the Analysis of Visual Accessibility Locomotion Techniques

To understand the techniques with the most potential for blind people, we investigated the analysed locomotion techniques more in-depth with the proposed categories (see section 3.2) and the UX attributes proposed in the *Locomotion Vault* [1]. We followed a filtering process where the filters were applied sequentially, where the output of the first filter was the input of the next filter. **In total, we applied eight filters**. Nonetheless, the filtering is commutative (the sequence does not affect the final result). The following table 3.6 shows the categories and UX attributes used in the filtering.

1 - *Vision*. In the filtering, there was no particular hierarchy or prioritization in the implementation of the aforementioned factors. However, since the work is focused on blind people, is natural to start with the most direct category (***Vision***), in this context, the filtering sequence started with the relevant categories of the prior work. After filtering the LTs that required ***Vision*** (from 107 to 69 LTs), the next step was to consider how the LTs contribute feedback information since visual feedback (blind people) is not an option to understand and navigate the virtual environment.

2 - *Output Type(s)*. Mindful that we are studying LTs in virtual reality, there is a modality feedback that can be taken advantage of - **proprioception** is a sensory modality that informs the body about its internal state, including awareness of the body's movements, position, and spatial orientation [63, 64]. Therefore, if a locomotion technique was analysed with ***Proprioception*** in the ***Output Type(s)*** category, it would pass to the next phase (from 69 to 58 LTs). In addition, other identified types can help blind people - ***Haptic*** and ***Audio***. However, with current guidelines and literature about blind and visually impaired people, these two types should be considered

mandatory, independently of the virtual environment interaction, in this context, both types were excluded from the filter, focusing more on movement information.

Table 3.6: Categories and UX attributes in the filtering process

Category	**Exclusion Value(s)**
Vision	*Required*
Output Type(s)	*Vision; Haptic; Audio*
Upper and Lower Limbs	*Indifferent*
UX Attribute	
"*Hardware*"	"*Board*"; "*Digitizer*"; "*Foot Controller*"; "*Pedals*"; "*Robots*"; "*Slip Surface*"; "*Specific*"; "*Treadmill*"; "*Wheel*"
"*Motion Direction*"	"*Vehicle*"
"*Spatial Awareness*"	"*Low*"; "*Medium*"
"*Intuitiveness*"	"*Low*"; "*Medium*"
"*State Of This Entry*"	"*Revise*"; "*Rename*"; "*Check Duplication*"

3 - *Upper and Lower Limbs*. With the established focus on body movement, the categories ***Upper*** and ***Lower Limbs*** fit the description. This filter is a conjunction of both categories (***Upper*** $\wedge$ ***Lower***), where the LTs had to be ***Required*** and ***Desirable*** in both (categories) to proceed to the next filter sequence. Typically, a disjunction (***Upper*** $\vee$ ***Lower***) would be ideal since both limbs contribute to proprioceptive feedback, where the LTs needed to be ***Required*** in just one of the two categories. However, when the filter was employed (disjunction statement) only one locomotion technique was filtered, rendering the filter unnecessary. Thus, the use of a more restrictive statement. Also, to ensure that the restriction is not causing over-filtering, was necessary to shorten the exclusion values, which ***Desirable*** was removed from exclusion properties (from 58 to 11 LTs). Considering that the remaining proposed accessibility categories have little impact on the filtering with a focus on movement, it was more relevant to analyse the UX attributes (Table 3.1) of the *Locomotion Vault* [1] than to continue pursuing the previous work.

4 - "*Hardware*". Mindful, that certain techniques rely on specific hardware, which in turn makes the techniques less accessible. We used the attribute "***Hardware***" from the *Locomotion Vault* [1] to remove LTs that require specific hardware (Table 3.6), besides the HMD and handheld controllers (from 11 to 8 LTs).

5 - "*Motion Direction*". According to the publication [1], "***Motion Direction***" is "*what determines the direction of movement (e.g., gaze, pointing, controller, walking)?*" [1], which the authors point out eleven types. Considering blind and visually impaired people, and the focus on movement in the filtering process, "***Vehicle***" could be a possibility, since for example, turning a wheel can provide movement information regarding the direction in which the arms were turned and positioned in the wheel. However, virtual vehicles even restricted by the system can cause dif-

ficulties for the blind user, for example, the auditory feedback to indicate their speed may not be accurate, causing a misperception of the distance the vehicle travelled in the virtual environment. Also, congenitally blind may have difficulty understanding the movement of a virtual vehicle if they have never driven a real one. Therefore, LTs analysed by the "***Vehicle***" attribute were excluded. Continuing "***Motion Direction***", there is a need to explain why "***Gaze***" is included in the filter, considering that the naming implies the use of the eyes. Based on the publication [1], the "***Gaze***" attribute is influenced by the study [65], where in "*gaze-directed steering, the user's view vector (typically the orientation of the head tracker) is used as the direction of motion, whereas the direction is obtained from the user's hand orientation in the pointing technique*" [65]. In other words, the "gaze" is not correlated to eye-tracking but to head movement (head tracking), which links with the focus on movement information, and could be a reliable element to help blind users to orientate in the virtual space (from 8 to 6 LTs).

6 - "*Spatial Awareness*". Following the order of the attributes in the *Locomotion Vault* [1], "***Spatial Awareness***" is another relevant attribute to consider - "*does the LT allow the user to be aware of their surrounding and their location and/or orientation in the VR environment?*" [1]. Considering the author's description of the attribute is relevant to include it in the filtering, where we want the techniques with high values. High spatial awareness is an essential characteristic that fits the target group considering the lack of visual ability. Therefore, the LTs with "***Low***" and "***Medium***" values were filtered (from 6 to 5 LTs).

7 - "*Intuitiveness*". When dealing with a specific technology (e.g. VR), there is a need to think about the training with the system. Because, not all users have experience in using the required devices, for example in [17], the participants needed eight hours to get used to the system and develop the required exploration skills since the process to accomplish an action, it's not the same in the physical world [66]. Thus, the attribute "***Intuitiveness***", where the authors [1] describe it as "*how easy is it to adopt this LT without any training or learning?*". Therefore, LTs classified with a "***High***" value are relevant to our work, excluding techniques that have "***Low***" and "***Medium***" values (from 5 to 4 LTs).

8 - "*State Of This Entry*". Lastly, it's relevant to consider "***State Of This Entry***". As the name implies, shows the accuracy of the recorded entry in the database [1]. It's relevant because ensures reliability in the information retained from the database of each locomotion technique. The database implements six states ("*Complete*", "*Good*", "*Preliminary*", "*Revise*", "*Rename*", and "*Check Duplication*") [1], and considering that there is no entry as "***Complete***" (time of this writing), there was a need to include the "***Preliminary***" ("*if they had not been fully implemented and evaluated*" [1]) to expand the scope of valid LTs. Thus, LTs analysed with "***Good***" and "***Preliminary***" states, were accepted in the filter, excluding the remaining, if existed. The remaining LTs after the filtering are as follows - Table 3.7.

Table 3.7: Promising locomotion techniques

ID	Name of the LT	State Of This Entry
08	"*Freedom Locomotion*"	Good
09	"*Natural Locomotion*"	Preliminary
38	"*Bobbing-Walking*"	Good
44	"*RIPMotion*"	Good

3.5.1 A Deeper Look into the Most Promising LTs

To further understand the potential of the abovementioned techniques (to be accessible), we evaluated the most promising techniques individually based on their features and functions.

"Freedom Locomotion". Based on the multimedia content in the database (e.g. LT creator presentation[7]), among other sources, such as the open marketplace for independent digital creators - *itch*[8] - "***Freedom Locomotion***" applies three LTs (*Controller Assisted On The Spot (CAOTS)*, *Dash Step* and *Blink Step*), as well as three always-on subsystems (*Grab Turning*, *Anti-Boundary Violation* and *Free Climbing*) that supplement the locomotion, making it more robust and intuitive. The concept is walking on the spot (Walking-in-Place), allowing multitasking, such as firing guns, while still moving around in the virtual world and looking in different directions.

- *CAOTS* tracks data from the bob of the HMD[9] combined with controller tracking and touchpad (LT creator used HTC Vive in the system presentation[7]) to control the direction of movement (to run, *CAOTS* accounts arm motion);
- *Dash Step* is an alternative of *CAOTS*, it uses short dash teleports that essentially mimic the action of taking a step, where the direction is controlled with controller tracking, and movement is with the click of a button;
- *Blink Step* is essentially the same as *Dash Step* but removes the visual motion between steps.

Two of the techniques described (*Dash Step* and *Blink Step*) are comparable to techniques within the *Locomotion Vault* [1] which were filtered previously, for this reason, both were not considered as potential LTs for the study. Furthermore, the three mentioned subsystems were also not considered.

- *Grab Turning* allows turning the virtual environment without having to physically turn. Since it does not apply the line of thought of using proprioception explained in the previous section (there is no need to physically turn), it can remove important information regarding the virtual space, disorientating the blind users;

[7]FL system Presentation - `https://youtu.be/LiQJS8ir9sQ` (Last visited April 11th, 2023).
[8]Itch - `https://hugerobot.itch.io/freedom-locomotion-vr` (Last visited April 11th, 2023).
[9]VR locomotion solved? The CAOTS system - `https://www.kitguru.net/components/vr/jon-martindale/vr-locomotion-solved-the-caots-system/` (Last visited April 11th, 2023).

- *Anti-Boundary Violation* allows the virtual environment to react appropriately to room-scale user movement, for example, by preventing them from walking through a wall or walking on air on a virtual edge. However, based on the video[7], the system relies on visual information;

- *Free Climbing* is a procedural climbing system, instead of designating surfaces or volumes as "grippable," the creator says it calculates the shape of the surface the avatar is gripping to determine whether or not it can climb on that surface. However, this system is comparable to an LT removed by the previous filtering (the *Climbing* technique). That said, the *Free Climbing* was also excluded.

Therefore, from the "***Freedom Locomotion***", *CAOTS* is the only feature/option that has accessible potential for blind people.

"*Natural Locomotion*". While in the *Locomotion Vault* [1] is described as a VR locomotion technique. "***Natural Locomotion***" is an application that allows changing the locomotion method [10]. "*The app allows the user to move more naturally in any game by emulating controller thumbstick input*"[11], all while moving the arms (arm swing) to walk or run, and jump using the bobbing of the HMD. At the time of this writing, the creator of the "***Natural Locomotion***" added new features that are not displayed in the video[10], these features are recorded in the *Steam* platform (video game distribution service) online page of the locomotion technique [11].

- *Feet Mode*, which uses motion sensor devices that track foot movement, where the user can choose between walking-in-place or swinging the feet while seated or when using a treadmill;

- And *Hands Mode* compatibility with body trackers, allowing for hip motion direction (the direction of movement is determined by orientating the hip in a certain direction).

Regardless, these two features were not considered, because even though a smartphone could be used as an alternative to track the data on both features (other more specific tracking hardware can be used, listed on the online page[11]), there was a locomotion technique that used a smartphone as the hardware to track movement data, which was excluded (filtered) in the previous section. Therefore, from the "***Natural Locomotion***" the *arm swinging to walk and run*, and the *headset bobbing for the jumping*, were potential options for blind people.

"*Bobbing-Walking*". As the name implies, bobbing the HMD allows moving forward in the virtual environment[12], however, we considered the technique redundant, because is comparable to *CAOTS* from "***Freedom Locomotion***", aside from using controller tracking and a touchpad to steer.

"*RIPMotion*". Like the previous locomotion technique (Bobbing-Walking) uses the bobbing of the HMD to move forward, however, the steering is with controller tracking as long they are

[10]NL system presentation - `https://youtu.be/qeoNQjoa5uU` (Last visited April 12th, 2023).

[11]Natural Locomotion on Steam - `https://store.steampowered.com/app/798810/Natural_Locomotion/` (Last visited on April 12th, 2023).

[12]BobbingBots - `https://youtu.be/RQmPODtp9Fw` (last visited on April 12th, 2023).

positioned at the hips[13], in other words, the avatar moves to where the body is pointing, meaning that turning the full body physically is necessary to navigate in the virtual environment. Considering the similarities with *CAOTS*, "***RIPMotion***" would be excluded, however, in the update presentation[13] the creator introduces the *LazyMode* feature which essentially removes the need to "jog in place" (which bobs the HMD), where the movement requirements are clicking a button and physically turn to steer the avatar. This difference can be significant to a blind user since the movement orientation is only provided by the turn, additionally, the input simplicity can be a positive element in the user descriptions. Therefore, from "***RIPMotion***" the *LazyMode* system was also a potential option for blind people. The following table 3.8 shows the options of the possible techniques that could be accessible to blind people.

Table 3.8: Potential features/options within the recorded LTs for blind people

ID	Locomotion Technique	Features/Options
08	"*Freedom Locomotion*"	*CAOTS*
09	"*Natural Locomotion*"	*Arm Swinging; Bobbing HMD to jump*
44	"*RIPMotion*"	*LazyMode*

3.6 Discussion

Making this accessibility catalogue allowed us to get a broader idea regarding the accessibility of the techniques. We found out that VR locomotion techniques within the *Locomotion Vault* [1] offer varying degrees of accessibility, catering to different impairments. For instance, we found out that the majority of the techniques use *Gestures* as an input modality. Therefore, even being challenged by the dependency of visual content in the techniques, blind people have possible alternatives since the physical interaction would provide additional feedback.

Next, we looked to see which of the techniques we could explore in our research for locomotion for blind people. To do this, we used the analysed techniques in the *Locomotion Vault* [1], to see which ones have more potential to be accessible, with a filtering process using three of the proposed categories and five of the UX attributes in the database (Table 3.6). In addition, we had a deeper look into the promising techniques, reaching the current techniques (Table 3.8). However, these LTs only range between a traditional option (*LazyMode*) and mimicking real-world movement (*CAOTS*, *Arm Swinging* and *Head bob to jump*). VR locomotion techniques can also be more versatile, representing out-of-the-box solutions (movements that cannot be accomplished in the real world) - e.g. teleportation (magical [1]). Furthermore, only two of the possible techniques (*CAOTS* and *LazyMode*) have different categories (Walking-based and Steering-based), while the third is a duplicate (Walking-based), not even representing half of the locomotion types mentioned in the literature [1] (see subsection 2.3.2). Therefore, we also looked at those (LTs) that are

[13]RIPMotion update - `https://youtu.be/IQ_jgnwHwFA` (Last visited on April 12th, 2023).

more common in virtual reality (e.g. Oculus VR Guidelines[14], [67]). We found out that the *Arm Swinging* feature/option within "*Natural Locomotion*", is a popular technique used in many VR applications.

With the knowledge of the techniques that have the most potential for blind people and a clear representation of locomotion available (and more prevalent) in common VR experiences (literature and market), we decided to implement three common locomotion techniques but different among them, which we contemplate further in the following section.

3.7 Limitations

Video-based analysis has several advantages, including cost-effectiveness, time efficiency, and flexibility, allowing access to a diverse pool of multimedia content anytime and anywhere. However, there are also limitations, such as the availability, and dependency of the information provided, for example, if the content is non-commentary is impossible to know user descriptions or, to understand and know the inputs the user provides to the VR system - e.g. recorded gameplay without user camera makes it hard to identify some inputs, where deduction is the main "tool". The same examples can be applied to the output (e.g. sound removal).

Additionally, other two variables in this study can be considered a limitation: the researcher's interpretation of the content and the attribute "*State Of This Entry*" within the database. Respectively, the exclusive evaluation of the techniques with description and visual content recorded in the database can make the catalogue results biased even with the same protocol, in other words, another researcher can have different results in some techniques. Another limitation is regarding the analysis to find the potential VR locomotion techniques for blind people. Some of the filtering steps are not just related to the accessibility of the technique but also to the ease of access - e.g. in "*Hardware*", we exclude techniques that rely on specific hardware, which does not mean that they are less accessible to blind people.

Furthermore, by the time of this writing the database entries could be modified or deleted. Therefore, this first study allowed an understanding of the current state of the *Locomotion Vault*[1] database in the context of a community-sourced tool, for a fully dependent search on the recorded information within the database, which currently still as some limitations - e.g. accessibility. Nonetheless, the database provides valuable insights regarding current locomotion techniques, and their functions and properties, giving clearer and faster access to information.

[14]VR Locomotion Design Guide - `https://developer.oculus.com/blog/now-available-vr-locomotion-design-guide/` (Last visited on September 14th, 2023).

Chapter 4

Implementation of VR Locomotion Techniques

In the previous study, we gained knowledge about VR locomotion techniques accessibility and which have the most potential for blind people within the locomotion vault [1]. To select techniques to investigate in more detail, we looked at those that seem the most promising based on our previous study, and those that are most popular (available in the market) and prevalent (e.g. [3]). Similarly to prior work [49, 68, 69] and our first study, we considered techniques that can be used in place (to avoid being constrained by the physical world) and that do not require any specialized hardware besides the headset and handheld controllers. In context, the *Arm Swinging* (within "*Natural Locomotion*") was an option that is promising, widely used in many VR applications and is considered a prevalent locomotion technique (Walking-based), which does not require full gait locomotion [3] (avoiding being constrained by the physical world). However, the last two are less common, with one among them (*CAOTS*) sharing the same locomotion characteristic as *Arm Swinging* (walking-based locomotion).

We selected techniques that are very different among themselves while also being popular in the literature and commercial VR applications - 1) **Arm Swinging** (Walking-based), which uses body movement (arms) similar to real-world walking; 2) **Linear Movement** (Steering-based), which relies on the controller's thumbstick; and 3) **Point & Teleport** (Selection-based), which instantly moves the avatar to a destination. These LTs, like the promising (Table 3.8) can be used in place (avoid real-world constraints) and maintain the original typing (Walking and Steering based), while adding a new one (Selection-based) to vary, and differentiate the VR experience, making sure we have represented the versatility of VR with techniques that provide a sense of realism or transcend the boundaries of physics (magical) [1], and represent the most prevalent types mentioned in the literature [3].

Following the aforementioned study, we proceeded to define the virtual environment with accessible interactions, in which the selected techniques would be implemented, and further contemplate the techniques, which we augmented each technique when necessary with haptic and auditory cues to ensure accessibility (e.g., the sound of footsteps and haptic/sound of collisions), as suggested by prior work (e.g., [9, 70, 71]),

4.1 VR Prototype

Our VR prototype was developed using Unity3D and the Meta Quest 2. The prototype was implemented with Unity3D native assets and the Oculus Integration package (Software Development Kit (SDK) developed by Meta for Unity). For the spatializer plugin, Oculus Spatializer was used with other assets that came within the package: Oculus Native Spatializer Plugin (ONSP) scripts, spatializer mixer example, and audio clips (e.g. from *RedGreenBall* and *LocomotionExamples*). In addition, we used *freesound* [1, 2, 3], *Mixkit* [4], *Pixabay* [5], *Unity asset store* [6], *Narakeet* [7], and *Firebase* [8].

4.1.1 Virtual Environments

We built two virtual environments: a training environment (180x180 meters) to introduce participants to the technique and to the VR application and a study environment (260x200 meters, with three possible layouts) to perform the study tasks.

Training Environment. This environment introduces new elements iteratively, allowing participants to learn new features gradually. It starts as an open, no-boundary space with two different types of terrain (Figure 4.1 a)), to understand how to move with a given technique and the feedback of footsteps (velocity and the sound of each terrain). Then, a wall is introduced to explain how collisions work (Figure 4.1 b)). Finally, an objective element producing sound is added to better explain spatialized audio (Figure 4.1 b)), and participants are tasked to reach it.

Study Environment. This environment features a boundary space, with grass terrain, and a gravel path with varying angles. Both the grass and gravel terrains include obstacles. This environment also features five objective elements that produce a looped piano sound [5], to convey the target location in space. These objectives are introduced sequentially (e.g., the second one only appears when the first is reached) and only one produces sound at each time. We designed three equivalent layouts (Figure 4.2) to be used in the study tasks (see subsection 5.3.1).

4.2 Locomotion Techniques

We aim to explore blind people's performance and preferences with the most common types of locomotion techniques – walking, steering, and selection-based [3]. Therefore, the selected locomotion techniques were augmented from the common design, meaning, that we maintained the core implementation of these techniques - the common/expected inputs (e.g. gestures, button

[1] Soft Tonal Wind - `https://freesound.org/s/397091/` (last visited on July 10th, 2023)
[2] Success Fanfare Trumpets - `https://freesound.org/s/456966/` (last visited on July 10th, 2023)
[3] Sad Trombone - `https://freesound.org/s/175409/` (last visited on July 10th, 2023)
[4] Wood hard hit - `https://mixkit.co/free-sound-effects/wood/` (last visited on July 10th, 2023)
[5] Chase Loop - `https://pixabay.com/sound-effects/67634/` (last visited on July 10th, 2023)
[6] Footsteps - `https://assetstore.unity.com/packages/audio/189879` (last visited on July 10th, 2023)
[7] Narakeet - `https://narakeet.com` (last visited on July 10th, 2023)
[8] Firebase - `https://firebase.google.com` (last visited on July 10th, 2023)

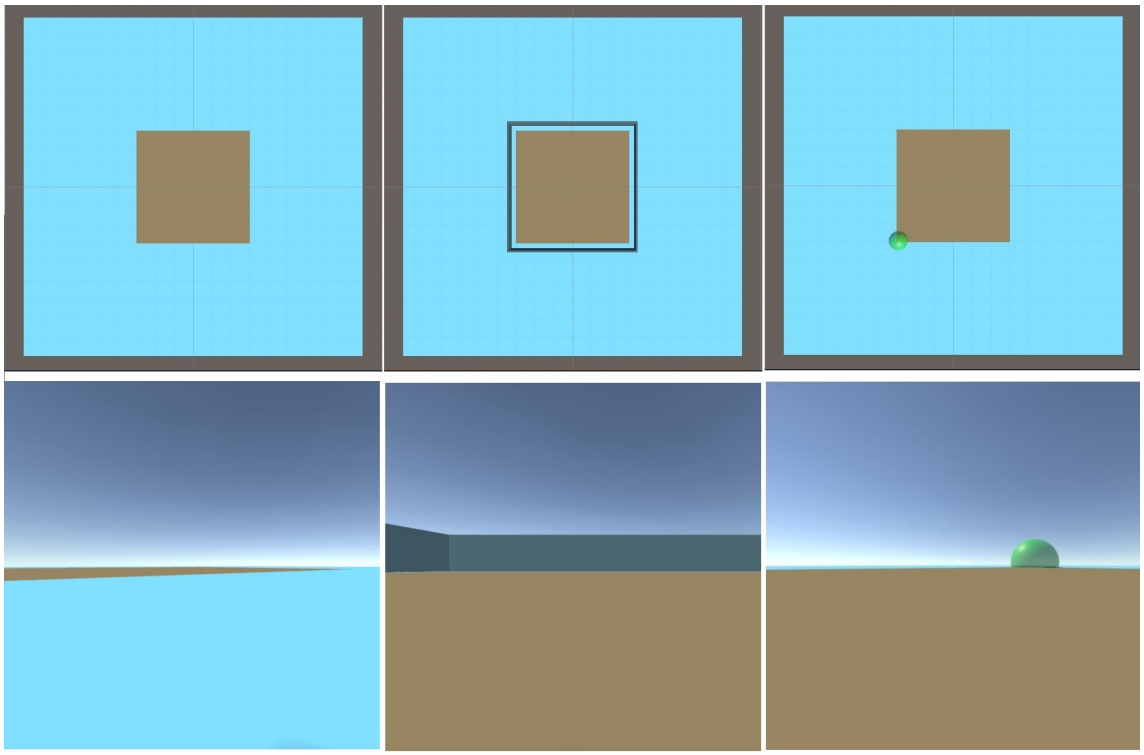


Figure 4.1: Training Environment with all the phases and 1st person perspectives: a) Phase 1 - Space with two terrains; b) Phase 2 - Introducing Walls; c) Phase 3 - Introducing Objective

clicks) that enable the technique in the application and augmented them with haptic and auditory cues (e.g., collisions represented with sound and vibrations) to support accessible navigation.

When possible, we used the native assets provided by Unity3D and the Oculus Integration package for the locomotion techniques. In the implementation of the cues, we performed preliminary testing within the research team and one pilot study with a blind participant, who helped refine our feedback cues.

4.2.1 Arm Swinging

Arm Swinging is a Walking-based technique relying on Partial-Gait locomotion [3], where users are required to move their arms while standing in place and pressing the *trigger* buttons (Figure 4.3 A1). The velocity of the virtual character depends on how fast users move their arms (Figure 4.3 A2), and the headset orientation gives the direction.

Among Walking-based techniques, Arm Swinging is the most explored in the literature [3] where the user stands in place and does not require additional hardware. In addition, it is included in many VR applications (e.g., Thief Simulator VR, Creed: Rise to Glory).

Since Arm Swinging is unavailable in the Oculus Integration package, we based our implementation on prior examples of this technique. In particular, the *Arm Swinger* technique[9] described

[9]Arm Swinger - `https://github.com/ElectricNightOwl/ArmSwinger` (last visited September 13th, 2023)

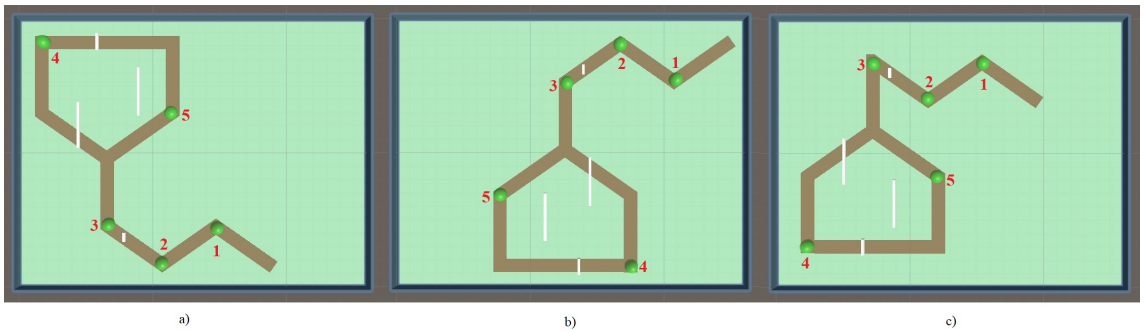


Figure 4.2: Study Environment with the three possible layouts, and all the objectives (and corresponding sequences): a) Layout A; b) Layout B; c) Layout C

in the *Locomotion Vault* [1], and a prior implementation described online [10]. We used the velocity of movement of the VR controllers, to understand how fast the user was moving. To ensure that the velocity aligns accurately with the user's perspective, the velocities (of each controller) were converted to world space where the maximum velocity is 2.5 m/s. This value was based both on prior studies that used 2 or 3 m/s [12, 72] and on experimentation to ensure realistic footstep rhythm.

The implemented additional cues to ensure the accessibility of this technique based on prior work on accessible virtual environments [9, 70, 71], include the sound of footsteps that indicate both the walking speed and the type of terrain. Collisions are announced with a bumping sound and the vibration of their controllers. For instance, if the user collides front-facing an obstacle, both controllers vibrate, but if the obstacle is on the right, only the right controller vibrates (and footsteps may continue if the collision – e.g., with a wall – is not stopping the user).

4.2.2 Linear Movement

Linear Movement (referred to as Joystick-directed Steering in [3]) is the most explored Steering-based locomotion technique in the literature [3] and is available in many VR applications. This technique relies on the controller's thumbstick to indicate the walking direction (e.g. with right and left corresponding to lateral movement) (Figure 4.3 B). Similarly to Arm Swinging, rotation is given by head(set) orientation, meaning the user can still rotate physically. Velocity depends on the force applied to the thumbstick, meaning that slightly pushing it forward would result in walking slowly. We used the implementation available in the Oculus Integration package. We changed the speed threshold to ensure a velocity range similar to Arm Swinging, with a maximum of 2.5 m/s.

The additional cues implemented in this technique include the sound of footsteps and collisions (as in Arm Swinging). In addition, we included a soft mono audio (wind sound) to indicate when the user is moving straight ahead (within a 10º radius). This was intended to prevent veering due to users slightly tilting the thumbstick to the left/right when trying to move forward, which was

[10] Unity (Walking In Place, Arm Swing) Tutorial - `https://youtu.be/Eipi6rNPz9U` (last visited September 13th 2023)

observed in preliminary testing.

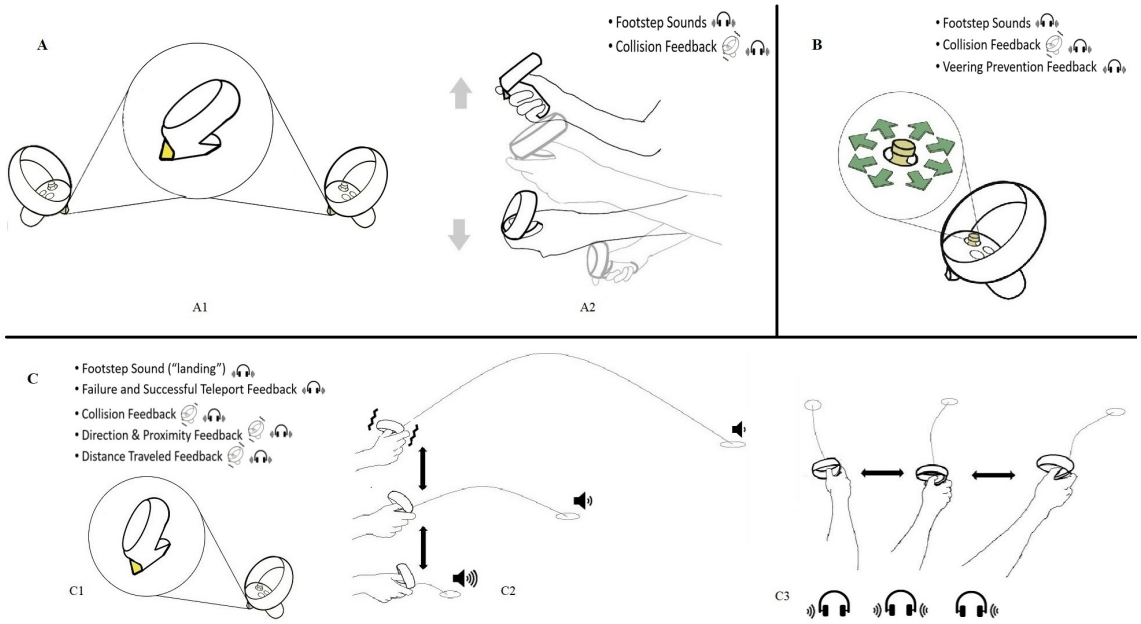


Figure 4.3: The three popular locomotion techniques with the respective feedback (the icons represent auditory and haptic feedback): (A) Arm Swinging - A1) VR controllers trigger buttons to engage the technique; A2) user moving the arms in an up and down motion; (B) Linear Movement input, displaying possible directions to push the thumbstick. The force applied to the thumbstick influences speed; (C) Point & Teleport - C1) VR controller trigger button to engage the technique; C2) user controlling the distance of the teleportation alternating the wrist upward to downward, vice versa; C3) user adjusting the direction of the destination.

4.2.3 Point & Teleport

Point & Teleport is the most explored Selection-based locomotion technique in the literature [3] and one of the most overall (both in the literature and commercial VR applications). Because of its ubiquity in different VR platforms (e.g. Meta Quest) [73], we used the arc-based or parabolic curve aim. This technique relies on the controller to point to a specific destination, instantly placing the user at the intended location after pressing and releasing the *trigger* button (Figure 4.3 C1)). In addition to the direction (Figure 4.3 C3)), the user can control the distance of the teleport by pointing the controller upward (further away) or downward (Figure 4.3 C2)). Head(set) orientation affects the audio feedback received (as if the avatar turns the head in the virtual world), but the controller's position and orientation are what dictate the direction of the teleport.

We implemented additional cues to support users in understanding the outcomes of their actions. To convey both direction and proximity, a sonar-like spatialized sound is emitted from the teleport destination when pointing to teleport. To help detect obstacles, a different sound is emitted (e.g., a bumping sound) when pointing to a location where teleportation is impossible (e.g., a wall). For confirmation, when teleport is successful, a distinctive "whoosh" sound is emitted. To provide additional feedback on the distance travelled, the length of such sound varies – meaning

longer the teleport, longer the "whoosh" sound, and vice-versa. After this sound, one footstep sound is emitted to convey the terrain where the user landed. When trying to teleport to an impossible location (e.g., a wall), a failure sound is emitted. Finally, a subtle vibration in the controller used for pointing is conveyed when reaching the maximum teleport amplitude.

We adapted the implementation from the Oculus Integration package to support the audio and haptic cues described above. The orientation of the controller indicates the direction of Teleport, and the maximum range is approximately 8 meters.

Chapter 5

User Study

We aimed to investigate how three common locomotion techniques support blind people navigating VR environments: **Arm Swinging**; **Linear Movement**; and **Point & Teleport** - with accessibility adaptations. We focus on widely popular techniques that lack the demanding prerequisites (e.g. specific hardware, or extensive/unrestrictive spaces), aiming to enhance our understanding of approaches more readily accessible to the general population. In particular, we wanted to answer the following research questions:

1. What are the differences among Arm Swinging, Linear Movement, and Point & Teleport in terms of performance and user preference?
2. What are the relative advantages of each technique, and how do users' interactions and strategies impact performance?

We conducted a user study with 14 blind people, where they were asked to complete VR navigation tasks using the three locomotion techniques. We evaluated the effectiveness (reaching the objectives) and efficiency (completion time and travel distance) of the three techniques in each scenario, performed a brief questionnaire, and conducted a semi-structured interview to gain a deeper understanding of user strategies and preferences. Furthermore, the Ethics Committee of our university approved the study.

5.1 Participants

We recruited 14 blind participants (Table 5.1) through a local training institution for people with visual impairments. Participants were between 26 and 63 years old (M=41.5; SD=10.4). Thirteen participants were blind or had light perception at most, while one (P11) had residual vision in one eye [1]. We excluded P11 from the quantitative analysis due to a higher level of residual vision, as well as P4, who completed only one of the tasks due to fatigue but considered their comments in the qualitative analysis. Most participants rated themselves as somehow experienced with technology but less experienced with virtual environments. Seven participants had never tried VR, while

[1] While recruitment was directed to people with no to very little residual vision, P11 reported having residual vision when starting the study, which enables to perceive some elements in the virtual environment.

six tried it once or twice. In particular, four participants experimented with VR in a prior research study, while the others did not specify their prior experience.

Table 5.1: Participant demographics, with experience (technology and virtual environments) rating from 1 (Not Experienced) to 7 (Very Experienced).

ID	Age	Gender	Technology	Virtual Environments	VR Experience
P1	63	M	4	4	1-2 times
P2	55	M	7	4	None
P3	38	F	6	4	None
P4	29	F	4	1	None
P5	39	M	5	1	1-2 times
P6	38	F	5	5	None
P7	26	F	7	5	None
P8	40	M	5	3	1-2 times
P9	49	F	4	3	None
P10	55	M	4	5	>5 times
P11	38	M	3	1	None
P12	35	M	5	3	1-2 times
P13	41	F	5	4	1-2 times
P14	35	M	6	4	1-2 times

5.2 Apparatus

We used the above VR prototype (see subsection 4.1.1), running on a laptop with compatible system graphics (NVIDIA GeForce RTX 3070). The wired connection gave more control over the experience, enabling the author to set/switch the environments and techniques as needed and have visual feedback. During the study tasks, participants were instructed to use headphones to enhance spatialized audio, while in the learning environment (see section 4.1.1), the audio was mostly communicated through the VR headset speakers, allowing for clear communication between the author and participant.

The study was conducted in a room at the training centre (P1-P13) or a location chosen by the participant (P14), ensuring enough space and low noise levels.

5.3 Methodology

The study had a within-subjects design, where all participants executed a navigation task with three different conditions – the three locomotion techniques. The order of each condition was counterbalanced, using the latin-square design. Our methodology revolves around quantitative analysis, involving participants' task performance and self-reported ratings (e.g., comfort, fun, efficiency), and qualitative analysis based on semi-structured interviews and observation.

5.3.1 Navigation Task

Participants performed three navigation tasks, one with each technique and layout (Figure 4.2). These tasks mimic traditional A to B navigation, which is common in many contexts where users know where they want to go (e.g., in Social VR or video games). We selected simple tasks (e.g., similarly to [74, 75]) due to the lack of knowledge about the accessibility of these techniques. For each task, participants were asked to reach five objectives as quickly as possible. These objectives emit a spatialized sound, enabling participants to understand their location. The objectives appear sequentially (from 1 to 5), meaning there is only one active objective at each time. An objective appears right after reaching the previous one or after a time limit of three minutes (for that objective) to avoid fatigue and frustration. If unsuccessful, a corresponding mono sound would play, and the avatar would be positioned on the objective location, ensuring all participants start from the same position when seeking the next objective.

We started with simple scenarios (from A to B, without obstacles or other users), moving to slightly more complex ones (e.g., including walls to cope with collisions). When starting, the avatar is misaligned with the first objective (90 degrees to the left or right) to requires reorientation (Figure 4.2). Objectives **O1** and **O2** only require reorienting and moving straight. **O3** requires moving straight but includes an obstacle in the middle of the gravel path. In these objectives, a gravel terrain represents the shortest path. While this may work as a hint, participants were not informed and were completely unaware of the environment, as we wanted to grasp if participants leveraged these differences. Objectives **O4** and **O5** increase the complexity of the task by having obstacles in the gravel path and along the shortest path. These intend to force collisions to better understand how each technique supports such scenarios. For that purpose, the obstacles are high enough to be impossible to pass over with Point & Teleport. To compare users' performance using each locomotion technique, we collected data regarding the number of objectives successfully reached, completion time and distance travelled.

After completing the task with each technique, we asked the Single Ease Question [76] where participants had to rate task ease/difficulty from 1 to 7 (1- Very Difficult, 7- Very easy). We also used 7-point Likert Items about the efficiency, fun, comfort, and accessibility of each technique (from 1- Strongly Disagree to 7- Strongly Agree).

5.3.2 Semi-Structured Interview

We conducted a semi-structured interview, focusing on understanding the rationale behind participants' preferences and each technique's relative pros and cons. In addition, we asked about their preferred way to navigate virtual environments in VR, and we tried to understand their perceptions of the accessibility of the environment and specific elements (e.g., collisions with walls). We audio-recorded the interview for later transcription and analysis.

5.3.3 Procedure

Sessions took on average 84 minutes (minimum 69, maximum 112). All sessions were conducted by the author and one or two additional researchers, where the author conducted the study, and the others observed, took their own notes and ensured the safety of the participant (e.g., moving the cable if needed).

Introduction. Each session started with a short overview of our research and its main goals. Participants were informed about their rights and signed a consent form, followed by a questionnaire focused on demographics and experience with technology and VR. The audio of the whole session was recorded after consent. Participants were then introduced to the VR equipment, where the author gave a brief explanation of its components while participants explored them with their hands to learn their size, weight of the headset and overall button position in the controllers. Then, participants were assisted in wearing the Meta Quest 2 hardware.

Study Trials. Participants were then asked to experiment with the three locomotion techniques in sequence. When starting with each technique, participants entered a training environment (see subsection 4.1.1), where features were introduced sequentially by the author, who provided detailed explanations about the feedback and interactions of each technique, followed by a Text-To-Speech (TTS) message from the system. After each message, participants could experiment with the technique and clarify questions with the author. After completing the training phase and making sure participants did not have doubts about the technique, the author started the navigation task, where a TTS message indicated the goal. After each task, we asked participants to rate the five 7-point questions and asked if participants wanted to take a short break.

Debriefing. After completing the tasks with all conditions, we asked participants to rank the locomotion techniques according to their preference and performed the semi-structured interview. Finally, we thanked participants for their time and insights. All participants received a gift voucher for their participation.

5.4 Data Analysis

We ran the Shapiro-Wilk test to assess the normality of the task performance metrics: number of objectives reached, completion time, and distance travelled. The completion time variable had a normal distribution for all techniques, while the remaining variables did not. To compare the three locomotion techniques, we used the ANOVA repeated-measures test for Completion Time, which involved a single factor (locomotion technique) with three levels, and the (non-parametric) Friedman test for the remaining variables. Similarly, we ran the Friedman test to compare the Single Ease Question and Likert Items results among the three techniques.

We transcribed all semi-structured interviews and conducted a mixed deductive-inductive thematic analysis, described as a codebook approach by Braun & Clarke [77]. The initial codebook was created based on our concepts of interest (for instance the three techniques, fun, efficiency, and accessibility) and our familiarity with the data complemented by our notes and observations

made during the study. Then, the author and the two additional researchers independently coded the same two interviews, adding new codes as necessary (e.g. orientation, real movement). They met to discuss the resulting codebooks to refine and ensure all relevant topics were captured. Then, the remaining interviews were split between the author and the two additional researchers. The themes presented in the findings result from multiple discussions and iterations over a live document with sense-checking by all researchers. Note that researchers translated participants' quotes into English as sessions were conducted in a non-English speaking country.

5.5 Results from Quantitative Analysis

We present our findings supported by a quantitative analysis of performance and participants' self-reported ratings and a qualitative analysis based on participants' feedback and observations of their navigation tasks.

5.5.1 Performance Analysis

There was no significant difference (p=0.17) in **the number of objectives reached**. Eleven participants reached all five objectives with Arm Swinging (M=4.92, SD=0.29), while 7 reached all with linear Movement (M=4.33, SD=0.98) and 8 with Point & Teleport (M=4.25, SD=1.48). One participant did not reach any objective with Point & Teleport, while on average, participants reached 13.50 objectives across the three techniques (and six reached all 15 objectives).

As for **completion time** (Figure 5.1), participants on average took less time with Arm Swinging (M=233.00, SD=87.89), but no significant differences (p=0.301) among techniques were found when compared with Linear Movement (M=297.50, SD=209.90) and Point & Teleport (M=320.67, SD=230.99). There were also no significant differences when comparing the time to complete each objective individually. Looking at the time spent on each objective, O5 was where participants took longer on all techniques, which was expected due to its complexity.

One participant reached all objectives with Point & Teleport within 46s, showcasing its potential for efficiency, while the minimum for the other two was close to 120s. On the other hand, Point & Teleport also represents the longest completion time (900s, for the participant unable to reach any objective with this technique), showcasing its greater complexity.

Finally, for the **distance travelled**, participants walked less with Arm Swinging (M=520.79, SD=143.56), with no statistically significant difference (p=0.097). The averages of the other two were close to 800m with high variance (Linear Movement M=765.84, SD=450.23; Point & Teleport M=837.42, SD=455.65). It was also with Arm Swinging that a participant walked the minimum of 414.17m, across all three.

5.5.2 Self-Reported Ratings

Through the debriefing, participants ranked their favourite techniques, showing a balanced result with varied preferences (Table 5.2). Linear Movement was the most preferred, but not by far (5

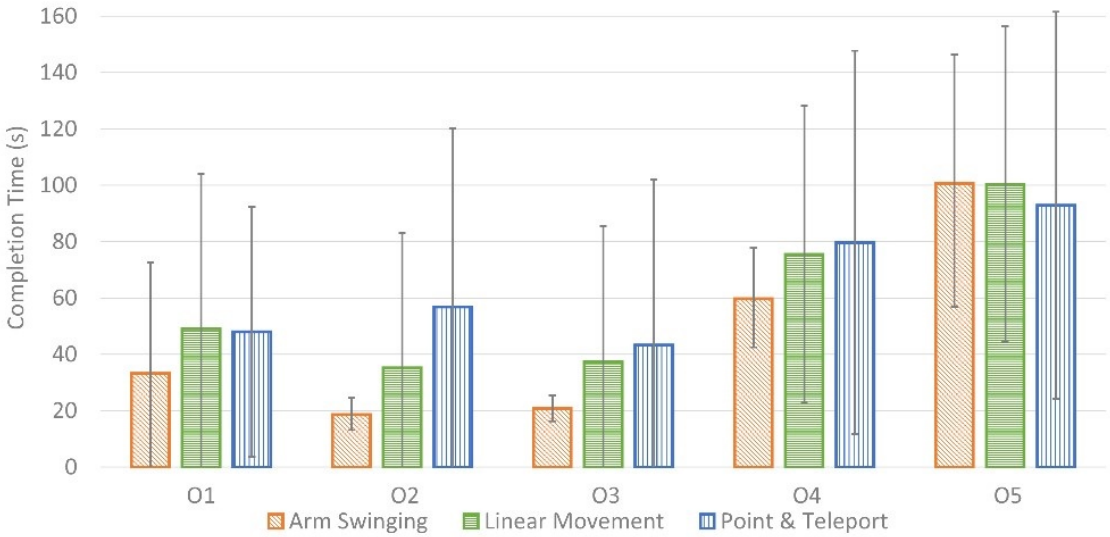


Figure 5.1: Average completion time per objective (in seconds) by all participants, for each technique. Error bars represent standard deviation.

participants preferred it, against 4 and 3 of Arm Swinging and Point & Teleport, respectively). On the other hand, Linear Movement was also the one most mentioned as least preferred (5 participants).

Table 5.2: Frequency of participants' order of preference for each locomotion technique

Locomotion Techniques	First	Second	Third
Arm Swinging	4	5	3
Linear Movement	5	2	5
Point & Teleport	3	5	4

After finishing each task, participants were asked to evaluate the respective technique according to its Ease of Use, Efficiency, Fun, Comfort, and Accessibility. We found no statistically significant differences among the three techniques in any parameter. Overall, the tasks were marked as **Easy** to complete by most participants for all techniques (Figure 5.2). In addition, all techniques had high, similar values for **Efficiency**, **Fun**, **Comfort** and **Accessibility**.

5.6 Findings from Qualitative Analysis

In this section, we present the findings divided into five main themes: preferences, dealing with direction, controlling movement, feedback cues and understanding the environment, and body language.

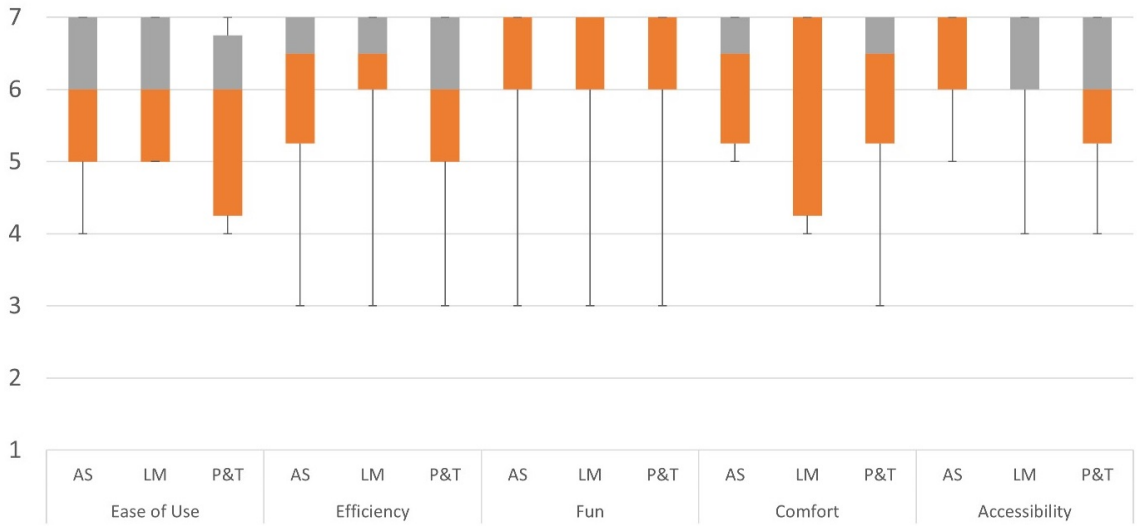


Figure 5.2: The Box Plot Graph demonstrates the variation of the answers from 1 to 7, according to the Likert Scale to different aspects of the Locomotion Techniques (AS - Arm Swinging; LM - Linear Movement; P&T - Point & Teleport) tested by the participants.

5.6.1 Preferences

Participants often justified their preferences when asked to justify their ranking. As reported above, preferences varied widely among participants as each technique significantly impacts the overall experience, by affording specific advantages for a subset of the participants. For instance, those very keen to exercise would like the physical activity of Arm Swinging, while others would prefer less tiring options.

> "*The arms require the person to move more and when the distances are long they have to swing their arms a lot*" - P1

It is this physical activation, however, the main factor for those preferring Arm Swinging, as stated by P3 and P14:

> "*It makes it more interactive and we are more aware of the movement (...) if we want to accelerate we move our arms, it's not something we have to press a button.*" - P13

> "*For me, for my gaming experience, I like it better when it is more physically demanding, let's say.. That it involves more movement, real.*" - P14

On the other hand, participants who enjoyed being physically active, described Linear Movement as uncomfortable or not as engaging:

> "*What I didn't like about the joystick was that you have to press a button all the time to move forward (...) it's a bit more uncomfortable.*" - P3

> "*I don't find it difficult at all, I just don't find it as stimulating, as much, as much fun*" - P14

Linear Movement was often preferred due to its simplicity and familiarity among occasional gamers. P1 stated that "*the joystick was the most fun [technique] and the easiest to learn*", while P8 referred to his familiarity with buttons in reference to gaming.

Conversely, participants felt that Point & Teleport was the hardest to master (e.g., "*But as I say, I had more doubts about using this technique than the others, but it went well.*" - P1). Its main advantages were its ability to move faster and afford a different experience:

> "*For a matter of . . . imagination (...) for an environment a bit different. And different in the sense that it is not natural, right? To human beings ... teleporting.*" - P14

Despite participants' preferences, the three techniques were generally rated positively and seen as appropriate for this setting:

> "*It's only because I have to choose one. Because for me they're all easy, all good. I liked them all.*" - P13

Other participants argued that techniques could be more appropriate for specific scenarios while sometimes also referring to possibly combining them depending on the challenge. For instance, P11 mentioned:

> "*In an entrance, I'm on the 2nd floor and want to teleport to the entrance of the building. Maybe it would be better for me to go downstairs in one click (...) but now, if I wanted to go to the office, it would be easier with my arm, to be sure of the path I'm taking*"

P14 agreed on the advantages of having all techniques available or even combining them in the context of a game:

> "*I think that the fact that I prefer the first one [Arm Swinging] doesn't mean that the player can't choose one of the three, and even go on to combine them. Let's suppose the game has levels. I may want to use a Teleport level in the first scenario because it is less difficult, so I can reach the objectives quickly. In the following levels, I may want to move more cautiously slowly.*"

5.6.2 Dealing with Direction

While Point & Teleport rely on the controller's position and orientation for direction, the remaining two techniques rely on the headset to convey body and head virtual orientation. All participants relied mostly on full-body rotation to align themselves with the objectives and then move forward in their direction. For participants, it was intuitive to use their bodies to face the intended targets, sometimes forgetting there were alternatives:

> "*As I was told that I would therefore walk forward when the joystick was in that direction, I always tried to use the movement of my body, I didn't even think.*" - P1

Interestingly, when rotating their bodies to try to locate the sound of the objective, four participants always rotated to the same side (e.g., only turning right) to understand/grasp the information. In addition, some participants performed occasional head (e.g. P9, P10, P11) or torso movements (e.g. P3, P11) to try to understand the objective's location.

Moving in other directions – other from forward – intentionally was very rare, even though both Linear Movement and Point & Teleport allowed movement in all directions without changing head orientation. The rare occasions included, for instance, P6 pointing to the side with Point & Teleport while maintaining fixed body orientation.

Unintentionally veering occurred multiple times and was the main reason for taking longer to reach the objectives. This was mostly caused by a mismatch of head orientation and input direction, sometimes leading to continuous circular patterns around the objective (Figure 5.3). Such behaviour was more common with Point & Teleport, as participants would slightly deflect the controllers when teleporting without realizing they were not pointing straight ahead. With Linear movement, the cause was usually a slight right/left movement with the thumbstick. In both cases, participants were often continuously moving and adjusting the orientation as they went, passing right next to the objective and then readjusting while moving. Participants, however, did not always understand what was happening in such scenarios.

"*There was that time when I heard the music, I'd go to the left side and then switch to the other side.*"

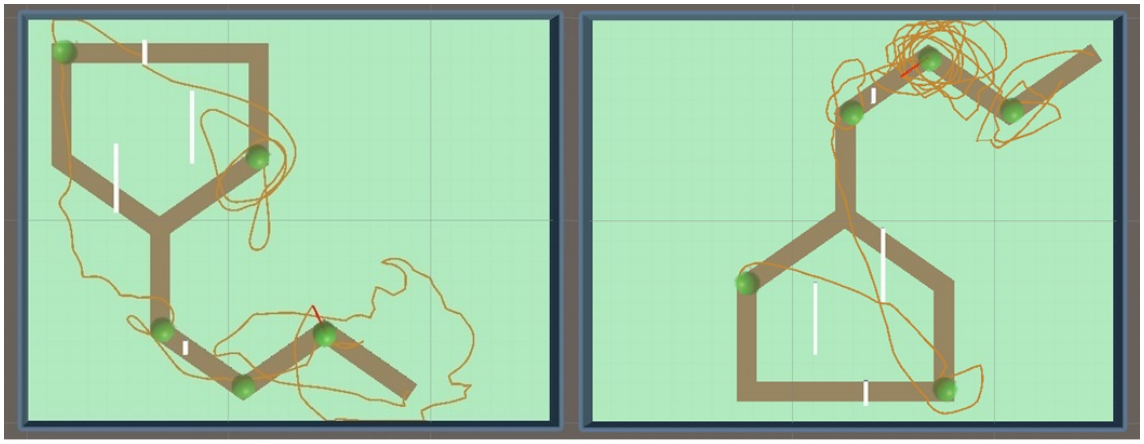


Figure 5.3: Top view of the environment with participants' trial trajectories, showing circular patterns - a) P9 (Linear Movement) on O5; and b) P6 (Point & Teleport) particularly on O2. Orange lines represent the path traveled by the participant and red lines indicate the automatic repositioning when the time limit was reached for a given objective.

The wind feedback was intended to prevent veering in Linear Movement, but the major focus on the objective led most participants to miss (or ignore) it:

"*I'll be honest, I didn't use it. I really felt it, but I was so focused on the goal that I didn't pay attention.*" - P1

Exceptions include P11 and P14, who referred to the importance of the wind feedback to prevent them from veering.

5.6.3 Controlling Movement

Each technique required specific interactions to control movement, which resulted in variations in performance and preference for each participant, despite the overall lack of significant differences in these criteria. The proprioception of Arm Swinging gave participants a feeling of being in movement:

"*The arm movement really is as if you were walking.*" - P5

"*It feels that I am really moving [when] using my arms.*" - P11

In addition, having greater control of movement may result in more precise interactions:

"*As the movement of the technique is slower (...) the arm technique allows for more constant control of the movement you are making.*" - P1

However, most participants did not vary the swing speed, keeping a constant (often the maximum) speed throughout the trial. This may be explained, in part, by the lack of feedback when the maximum speed is reached, meaning participants could vigorously swing their arms without producing differences in speed:

"*In this technique, I didn't know the maximum speed. That regardless of swinging my arms quickly, the speed wouldn't increase*" - P1

Overall, the style and speed of swinging varied among participants with some vigorously swinging their arms, and others swinging them gently, resembling a soft jog. Still, both P1 and P14 referred to Arm Swinging as slower than the others (despite having the same maximum speed as Linear Movement) and suggested increasing the maximum speed when Swinging vigorously.

With Linear Movement, participants usually moved forward at maximum speed, as they could easily push the thumbstick until its limit. This resulted in a simpler interaction that was valued by participants (e.g., "*It's just your fingers, the button on the side, to the left and up, it didn't take much work*" - P12). Still, it also provided less control and notion of speed (P11, P14), contrasting with Arm Swinging:

"*I know that it's all .. down, at maximum speed but I don't feel the speed itself, it's as if there's no movement that would lead me to conclude that I'm really at maximum speed*" - P11

Alternatively, a few participants controlled movement by giving small pushes on the thumbstick, resembling clicking a button and assessing their next move continuously. In addition, participants referred to the freedom of movement given by the thumbstick, which resembles locomotion in video games:

"*Is the movement of the joystick. It's that thing where you can have freedom... (...) The joystick allows you to move from side to side, make diagonals, play... with the game, you know?*" - P8

However, most participants only pushed the thumbstick forward, with very few occurrences of lateral walking. Just pushing forward lets participants focus only on body rotation to reach the objectives, which may also cause a feeling of observer as pointed by P1 (whose favorite technique was Linear Movement):

"*It's curious because while in the others I was aware that I was moving, when I was using the joystick technique I seemed to be more concerned with directing myself towards the goal and more 'in that goalkeeper position waiting for the ball to arrive'.*"

Point & Teleport differs from the others in the lack of continuity of movement, which is impossible to replicate in the real world. This caused uncertainty in understanding how to interact with this technique, even for those who performed the best:

"*But as I say, I had more doubts about using this technique than the others, but it went well.*" - P1

In addition, this technique included additional auditory cues, which may have raised unique barriers to participants. P6 referred to greater difficulties using this technique:

"*I don't know if I'd say it was a disadvantage, but I sometimes had a hard time getting to the sound.*"

Some participants referred to Point & Teleport as the most efficient:

"*It's faster. In one click you're right where you want to be (...) And at the same time, you can make the jump to return, which means I can jump to the other side and back in two clicks*" - P11

Still, interactions with this technique were different among participants. The most efficient ones usually performed "consecutive jumps", pressing quickly to teleport, while making minor orientation adjustments. P1 was the most extreme case as he took 46 seconds to complete the navigation task with Point & Teleport, while the fastest trials with Arm Swinging (122s, P3) and Linear Movement (116s, P1) were much slower. Other participants, would reassess the auditory feedback of the objective after teleporting, taking longer to complete the trials. Participants also varied their use of teleport amplitude and scanned the surroundings. While some participants always used maximum amplitude, others reduced the length of their teleport when getting closer to the objective, showing awareness of proximity and fear of over-shooting.

5.6.4 Feedback Cues and Understanding the Environment

The environment and the techniques have feedback mechanisms encompassing different audio and haptic cues. All participants agreed on the accessibility of the environment and of the techniques, showing that minor adjustments and careful (but simple) audio and haptic design enable blind people to use these popular techniques. Overall, participants were aware of all elements in the environment, as portrayed by P14's comment:

> "*In terms of terrains, it had three types, as I understood, clay, grass, and water. Ahh, then we had the walls, which were perceived with the vibration of the controllers and with the sounds of .. beating, almost percussive. And in terms of sound, we had a human voice singing and a piano. And in the third [Linear Movement], we had the wind, the sound of wind to help with orientation.*"

The spatialization of sound was important for participants to understand the direction and proximity to the objective. The sound of footsteps was relevant to understanding movement and speed, but the different terrains, despite being noticed, were somehow ignored by participants, who did not grasp their potential relevance to the task.

Multimodality was valued on collision feedback (vibrations and bumping sounds), as the vibrations gave an additional cue to what was happening:

> "*The question of vibration was very important when we hit the wall, because it's different information, there you go, when we're listening to the music, we're listening to the footsteps, the wind (...) we can see that something strange has happened there.*" - P3

All participants tried to identify the objective's location and moved toward it except when blocked by obstacles. Herein, the locomotion techniques afford different strategies to surpass them. With Arm Swinging and Linear Movement, participants often walked alongside the wall using it as a guide until there was an opening to the objective (Figure 5.4, a) and b)).

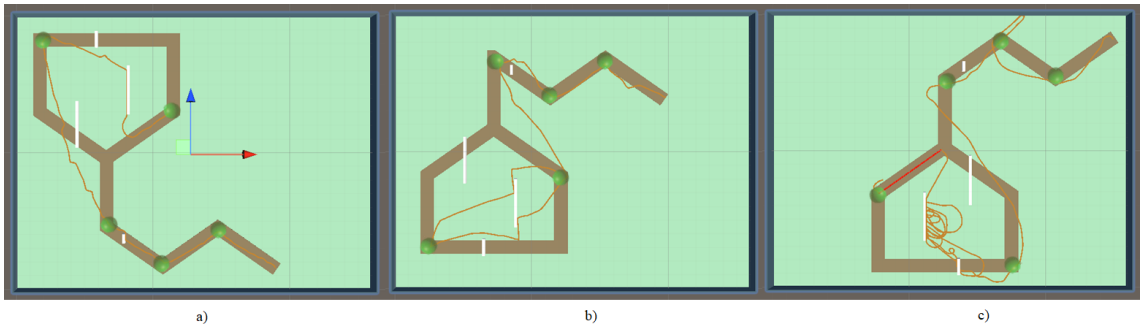


Figure 5.4: Top view of the environment with participants' trial trajectories - participants using obstacles as a guide: a) P1 (Arm Swinging); b) P14 (Linear Movement); and c) P7 not finding the limit of the obstacle (Linear Movement). Red lines represent the repositioning of the avatar after passing the time limit.

Still, as there is no length indication, participants often felt uncertain about the best strategy to surpass the obstacles:

> "*I couldn't understand at first if the walls were continuous, or if they were just... a sort of barrier... well, shorter, with a beginning and an end. No, I couldn't understand that and I had some difficulty.*" - P14

In such cases, some participants tried to get further from the obstacle and then tried again, but sometimes circled and went back to the same obstacle (Figure 5.4, c)).

Point & Teleport afforded a different strategy as participants could sweep their arms and scan for obstacles to understand when there is none anymore. This allowed participants to bypass obstacles and make efficient trajectories (Figure 5.5). Still, when too close to an obstacle, participants' trajectories would often resemble those of the other techniques as participants were required to go around it.

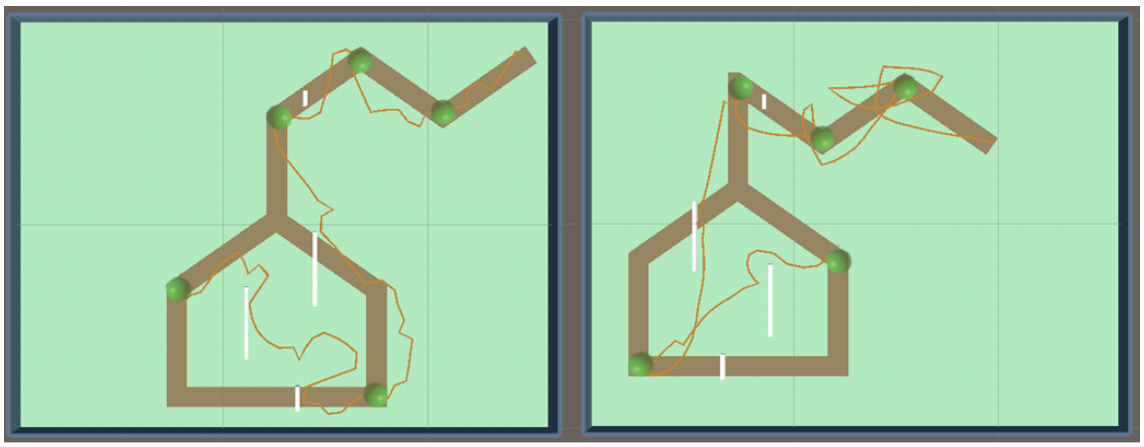


Figure 5.5: Top view of the environment with participants' trial trajectories - Example of Point & Teleport strategies, participants going around the last obstacle: a) P10; and b) P13

Participant's focus on the objectives sometimes led to confusion as in specific cases some felt overwhelmed when dealing with the last obstacle:

> "*I knew I had to move, but there was the noise of hitting the wall and the noise of the music and I was like, well and then I suddenly just (...), 'you have to go around because you're hitting the wall'.*" - P3

This can be explained, in part, by the lack of occlusion in the environment, where the obstacle in front was not occluding the objective sound. In such cases, a few participants ignored the collision feedback, describing that the sound was right in front of them:

> "*I was so focused on the sound that it seemed like the sound was already there... That I'd already located the sound and it was right in front of me (...) but at the same time I could hear the noise of the wall and the remote vibrating a lot.*" - P11

In extreme cases, participants maintained the same behaviours and continued colliding with the obstacle until the time limit was reached. When asked how each technique supports understanding the environment, most participants considered all equally able. Arm Swinging was sometimes considered the one with the greatest potential due to proprioceptive feedback. On the other hand, P1 and P11 mentioned that the lack of continuity in Point & Teleport can bring new challenges in gaining awareness of the space travelled and mapping the environment:

> "*I'm going to a space, but I don't know exactly which one, it's not me making the path, it's as if I'm appearing and disappearing.*" - P11

> "*[With Teleport] the person goes from one place to the other, but the space traversed by the jump ceases to be explored*" - P1

A few participants occasionally deviated from the objective, leaving the range where the sound is audible. In such cases, participants continued an aimless route until finding an obstacle e.g., the walls limiting the environment - (Figure 5.6) and then reorient their trajectory in an attempt to get closer to the objective.

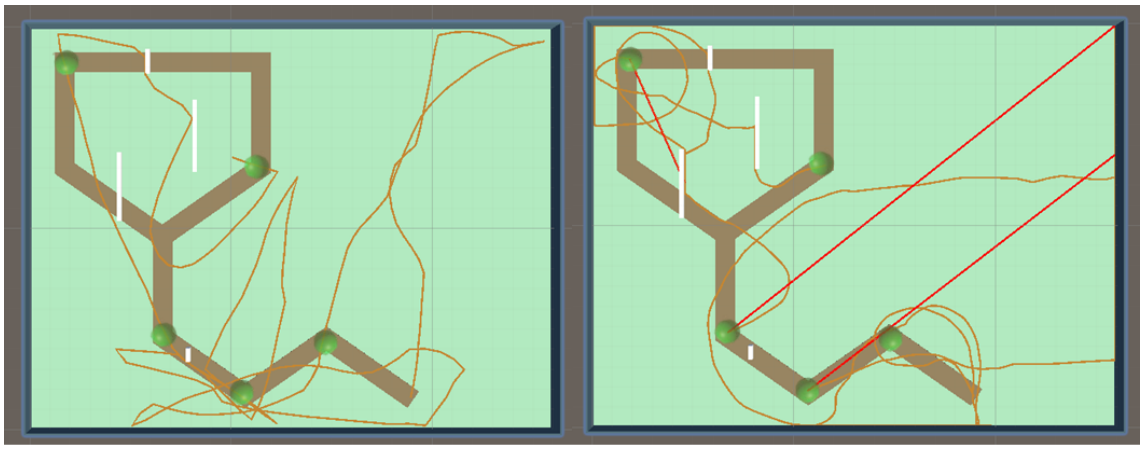


Figure 5.6: Top view of the environment with participants' trial trajectories - Example of aimless exploration: a) P8 reorienting trajectory (Point & Teleport); and b) P2 using wall to explore (Linear Movement). Red lines represent the repositioning of the avatar after passing the time limit.

Overall, and despite being aware of all the feedback cues, participants' focus on the objectives resulted in little knowledge about the spatial arrangement of the environment. One exception was P3, who described the three layouts as identical despite their subtle differences:

> "*The environments of the three techniques were identical. They started with a straight, very simple route, then, if I'm not mistaken, a more diagonal one, and a third that you had to turn to the left, and always go more or less straight ahead.*"

5.6.5 Body Language

We explained how to interact with the VR application and hardware, ensuring participants could perform the actions required comfortably and effectively. Still, we did not force a specific posture or way of interacting with the system. This led to differences in how participants controlled their movement, especially in Arm Swinging where the "swings" varied from pulling (e.g. front to back), using circular motions, or using only one arm up and down. In addition, P14 performed as instructed but suggested a motion close to cycling:

> "*Instead of up/down, it could perhaps be a different technique: front/back (...) In fact, it could even combine the two issues, which is, this almost rotational movement (...) which, in the end, is probably even closer to walking (...) so maybe this is a more bicycle-like movement.*"

In Arm Swinging, participants also unconsciously started walking in the real world, expressing difficulties in suppressing this movement when first introduced to the technique. P1 referred to this tendency to move, especially with Arm Swinging, but also with the other techniques.

> "*The most challenging thing for me was really being able to control the movement of my body, specifically to direct myself in the right directions and then even with the other techniques, I was able to inhibit that tendency to move forward, I think control of position is important.*"

Participants also held the controllers in different ways. For instance, some participants kept them close to their chest in all techniques, as a way to prevent accidental collisions with real-world objects. Still, some hit both controllers against each other or even with the headset (especially with Arm Swinging).

Point & Teleport has also created unexpected behaviours, as many participants would sweep their surroundings (especially facing forward), sometimes resembling a white cane. P7 actually referred to its similarity with a white cane during the learning phase. P6, on the other hand, would keep a relatively stable head orientation while teleporting in the direction of the sound, which contrasted with the body posture of the other participants who would try to align their heads with the objective.

Linear Movement observed a similar posture among participants, independently of how they held the controller (e.g., either closer or further their bodies). An exception is P9 who would actually point the controllers in the direction of where she intended to move.

5.7 Discussion

We investigated if and how three popular locomotion techniques support blind people navigating VR environments. Results show that most participants could complete the navigation tasks successfully, suggesting that the core implementation of these techniques augmented with careful but simple audio and haptic design (e.g., using footsteps, collisions, and teleport feedback) enables

blind people to navigate in VR experiences. In this section, we answer our research questions by summarizing the main results of the user study, and we discuss the lessons learned from our analysis, aiming to inform the design of accessible VR experiences and locomotion techniques.

RQ1. What are the differences among Arm Swinging, Linear Movement, and Point & Teleport in terms of performance and user preference?

Our quantitative analysis revealed no significant differences among techniques in the performance metrics - number of objectives reached, completion time, and distance travelled. These results differ from the participants' subjective ratings in *BlindWalkVR* [51], who perceived joystick-based interaction as more precise than three walking-based techniques (that rely on treadmills or feet trackers), and faster (along the Omni Treadmill) than the others. In addition, prior studies with sighted participants report faster completion times with Teleport- than with Joystick-based techniques [78]. However, our study revealed that participants' performance was highly user-dependent as, for instance, Point & Teleport resulted in both the fastest and the lowest trials.

Similarly, preferences were balanced across users and were often aligned with performance – but not always (e.g., P1 preferred Linear Movement, but was much faster with Point & Teleport). Regarding Ease of Use, Efficiency, Fun, Comfort, and Accessibility, participants' self-reported ratings showcase very positive ratings but, again, balanced across techniques. Prior comparisons performed with sighted people found differences in related aspects; for instance, Joystick and Arm Cycling (similar to Linear Movement and Arm Swinging, respectively) were classified as more enjoyable than Teleport (among others) [69]. Another study reported an overall preference for Teleport (and redirected walking) over joystick-based interaction [49]. In contrast, our study showed divergent, but balanced preferences among blind participants, while all conditions were perceived as equally fun.

RQ2. What are the relative advantages of each technique and how do users' interactions and strategies impact performance?

Each locomotion technique afforded specific interactions, providing different advantages to participants. For instance, the proprioceptive feedback of Arm Swinging provided greater control over mobility, while Linear Movement was mostly praised for its familiarity and simplicity. On the other hand, Point & Teleport has the potential for efficiency and for inspecting the environment.

These advantages promoted different strategies by some participants (e.g., scanning for obstacles with Point & Teleport, and trailing the wall – a common strategy in the real world [79] – to surpass it with the others), which impacted overall performance.

5.7.1 Lessons Learned

We extend the answers to our research questions by discussing the lessons learned from the findings of the user study.

Arm Swinging, Linear Movement, and Point & Teleport can support accessible VR Experiences (with appropriate audio and haptic design). The three techniques were augmented with audio and haptic cues that provide crucial feedback for users to perceive their movement and the surroundings. Feedback of footsteps, collisions, different terrains, or to indicate the objectives have been suggested or implemented in the literature related to virtual environments for blind people [9, 51]. Findings have shown participants' ability to (almost always) reach the objectives with all techniques in a relatively simple task. VR designers may use these findings to select and implement popular locomotion techniques that support accessible experiences with little to no additional effort – e.g., many experiences already include a sound for collisions and footsteps – and leverage our adaptations as a starting point for their techniques and experiences. One may speculate that similar techniques (e.g., other walking-based techniques) would also result in overall positive performances, meaning that – with careful audio and haptic design – most locomotion techniques available can support accessible VR experiences for blind people.

That being said, further research is needed to understand how each technique supports different scenarios, especially if considering more complex tasks and experiences. For instance, prior work has investigated differences in spatial understanding, suggesting that a VR treadmill was more effective in remembering routes but less effective in remembering obstacles than VR trackers (walking-in-place) [52]. In our study, participants referred that while Point & Teleport could be used to quickly reach a destination, Arm Swinging could be used when more caution or precision is needed. Experiments under different scenarios and complexities can help further understand the pros and cons of each technique.

Body Rotation makes interactions even simpler. Participants took advantage of the proprioceptive feedback afforded by VR and the locomotion techniques. Body rotation is important for spatial orientation [63, 64] and prevailed over other orientation techniques, such as rotating the head alone, or moving sideways with Linear Movement (or Point & Teleport), as participants usually moved straight ahead. This was more evident in Linear Movement due to its simpler and more familiar interaction with the thumbstick [51], which was further simplified by participants who relied on body rotation to deal with direction, using the thumbstick only to move forward – or diagonal-forward, sometimes unintentionally (veering). Such behaviour suggests VR designers may rely on body rotation as the main orientation factor, especially in Linear Movement where participants' mode of interaction questions the need for a typical thumbstick for input – in such cases a single button for navigation could suffice, complemented by head orientation to steer (e.g., similarly to the "*LazyMode*" in the "*RIPMotion*" technique [1]). Still, designers should consider that such simplification would affect the freedom of movement that would allow other types of interactions.

Arm Swinging is perceived to provide greater awareness and control of movement. Arm Swinging, as a walking-based technique, adds the proprioceptive feedback of walking by swinging the arms. This technique resulted in mixed opinions regarding the physical activity required — those keen to it enjoyed it, while others found it tiring – a fact also observed in the literature [12].

Some participants reported having greater awareness and control of movement with Arm Swinging, despite the lack of differences in overall performance metrics. Also, most participants moved at a constant, maximum speed (which also happened with Linear Movement). This suggests that their perceived control may not have been reflected in the virtual environment, likely due to a low maximum speed threshold. Designers and future research may increase such values to provide greater speed variations, potentially improving actual control over movement. However, despite the lack of reports of cybersickness in VR for blind people, special attention must be given to the sound of footsteps to keep them natural and accurately convey speed.

In addition, our tasks and metrics did not focus on participants' spatial representations of the environment, but anecdotal evidence suggests loss of information when using Point & Teleport since this space is not travelled, and a reduced perception of speed when using Linear Movement. While this suggests that designers may use walking-based techniques when such spatial representation is important (as also hinted in [49]), further research is needed to evaluate how each technique supports building mental representations of VR environments.

Point & Teleport has great potential for efficiency but may require additional training. Prior studies on VR locomotion for blind people have focused on continuous walking through the virtual environment with either walking- or steering-based techniques [8, 51, 52]. To our knowledge, this is the first work studying Selection-based (specifically, Point & Teleport) with blind people and comparing it with other techniques. This technique contrasts with the automated ones sometimes found in digital games, which impact users' sense of agency [9], but enables users to quickly move to a desired destination within a specific range, in any direction. Some participants leveraged this by performing rapid consecutive teleports until reaching the destination. P1 spent less than half the time with this technique than the fastest participants in the other two techniques. This links to findings of prior studies with sighted people [48, 78].

Despite its potential, findings showed that Point & Teleport was the most efficient technique only for a small subset of participants. This was mostly caused by its greater complexity, which is also a drawback for sighted people [73, 80] who benefit from experience with the technique [78]. Designers may leverage this knowledge to create mechanisms (e.g., through practice or tutorials) to speed up learning when needed. In addition, Point & Teleport required additional feedback cues to give a sense of distance. Future research may seek ways to reduce complexity while maintaining its advantages. In addition, it may investigate how blind people's continuous experience with this technique (and the others) impacts performance.

Point & Teleport can be designed to support scanning the surroundings. Point & Teleport offered the unique option to scan the surroundings without moving in the virtual environment. In our study, this feature could detect obstacles and assess their distance, supporting participants' decision-making through more efficient trajectories around obstacles. A parallel can be made with spatial awareness tools developed in the contexts of digital gaming [23], in particular, NavStick [11], which allows blind gamers to inspect what is around them.

This feature may gain further relevance in more complex environments where the user's avatar

may incur damage (or worse) by moving to a specific location (e.g., to a hole, cliff, or lava terrain). In such cases, upfront feedback may give further knowledge about the environment, which may provide alternatives to preventive strategies (such as constantly saving the game to load in case of hazard [9]) or fail-safe interactions [81], which decrease the challenge and sometimes user's agency. Further research may explore how this kind of feedback can be provided to blind users in walking-based techniques (e.g., as a preview), as the terrain (or its absence) is only perceived after stepping on it.

Specific body language may affect performance. We observed differences in body language linked to the Arm Swinging technique, including a suggestion to perform rotational movements resembling riding a bicycle (which is an existing technique [69]). These differences in body language may be related to blind people's use of the white cane or guide dog, which may prevent this movement in their daily lives or to the absence of prior examples (e.g., by observing others or videos) that are usually presented visually (as it happens, for instance, with gestures [82]). Designers should consider these interactions by ensuring they are accounted for or providing instructions on how to execute them. Another impact of specific body language is the frequent veering, especially in Point & Teleport, where some participants struggled to point the controller straight ahead, negatively influencing their performance as further discussed below.

Veering is also a problem in virtual navigation. Veering was the cause of most cases where participants missed an objective or circled it repeatedly. Veering is also common in real-world locomotion [83, 84], and researchers have proposed systems to either prevent or correct it [15, 85]. In this study, we included a subtle sound in Linear Movement to indicate participants were walking straight, but most participants ignored these due to being focused on the objectives. Future research may explore if more noticeable corrective sounds have a greater impact in preventing veering while ensuring the experience is not negatively impacted.

We observed that a mismatch between head and input (controllers or the thumbstick) orientation was what often caused veering, both in Point & Teleport and in Linear Movement. In the first, participants would often point the controller with a slight diagonal orientation, while in the latter, the same happened with the thumbstick. In contrast, Arm Swinging relies on head orientation to control direction, causing less veering. This suggests that techniques with movement restrictions – e.g., relying on the head alone to control direction [68] – may prevent veering. Still, this comes at the cost of freedom of movement, since users can only move forward.

5.7.2 Limitations

Numerous locomotion techniques have been proposed and implemented, each with some possible variations. In our implementations, we tried to use representative examples of each technique but were required to make decisions that may impact the results. For instance, the decision on maximum speed was defined based on prior work and experimentation, but there is no standard available. Faster locomotion could, for instance, be perceived as unnatural or influence the performance results. In addition, the haptic and audio feedback decisions may influence the results but

were needed to convey information about what was happening in the environment. Nonetheless, we carefully introduced feedback based on prior research and current practices. For instance, footsteps and collision feedback are common in virtual environments, and we used spatialized audio to convey the location and distance of the teleports.

The tasks and environment used allowed us to understand how these techniques support accessible navigation in simple scenarios, but VR applications are often more complex and dynamic, with multiple objects and users. Further research is needed to understand if and how these techniques should be adapted to cope with the additional information and feedback in such scenarios.

We also note that the study included a relatively small sample of diverse participants, which likely contributed to highly user-dependent results and consequently a lack of significant differences in the performance metrics. While we did not observe personal characteristics that could have influenced these performance differences, future research may explore what is impacting such diverse results. On the other hand, participant diversity enabled us to learn from a broader set of experiences that resulted in rich qualitative findings.

Chapter 6

Conclusion

Prior research has made significant contributions, frequently relying on custom-made solutions that demand specialized equipment or extensive/unrestrictive spaces. Unfortunately, this specificity often restricts the widespread adoption and implementation of such techniques due to their intricate setup requirements. In this work, we focused on understanding the potential of current techniques, assessing their accessibility, and on widely popular techniques that lack the demanding prerequisites (e.g., specific equipment), aimed to enhance our understanding of approaches more readily accessible to the general population.

We investigated the accessibility of VR locomotion techniques, and for that, we first augmented the *Locomotion Vault* [1] to gain more knowledge about VR locomotion techniques accessibility, which we propose 14 categories. We found that, while *Gestures* continue to be the most used modality, *Hand Controller* (or traditional interactions), follow closely with very little difference. In addition, even though VR locomotion allows for full-body movements, such as jumping, our catalogue shows that the majority of techniques are indifferent to lower limbs. Finally, while was expected to see high dependency on *Vision*, there was little to none regarding the remaining senses (e.g. *Speech*, or *Hearing*). Then, we looked and selected potential locomotion techniques, based on both the literature and resources available on the market.

We implemented three popular VR locomotion techniques and augmented them with auditory and haptic cues for accessibility. We performed a user study aiming to compare the three techniques in terms of performance and preferences, as well as to understand their relative advantages and the strategies they support. Results showed no significant differences in performance but contrasting preferences where both were highly user-dependent. Ultimately, they also showed that popular locomotion techniques prevalent in mainstream VR applications are accessible to blind people, as long as appropriate audio and haptic cues are included. If considered by design, these cues may pave the way for more accessible VR experiences, while the lessons learned from our work may inform the design of locomotion techniques that improve blind people's experience and performance.

6.1 Future Work

Regarding future work, there are multiple directions to take. For instance, our work showed, that not all natural modalities are being used in current VR locomotion techniques - e.g. voice commands. While the feedback needs to be carefully thought out and designed, there is a chance to explore other VR locomotion techniques based on voice input, maintaining an immersive environment with ambient sound and other engaging elements, allowing for an accessible interaction with blind people.

Another promising path is to investigate the techniques in more complex environments. Further research is needed to understand how each technique supports different scenarios, especially if considering more complex tasks and experiences.

An additional direction worth exploring is a work focused on mental mapping-oriented tasks. Where studying the cognitive mapping of blind people using, for example, the three implemented techniques, supporting and contributing further O&M training.

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